

Analog Design and Machine Learning: A Review

Konstantinos G. Liakos *  and Fotis Plessas

Department of Electrical and Computer Engineering, University of Thessaly, Sekeri Street, 38334 Volos, Greece; fplessas@uth.gr

* Correspondence: kliakos@uth.gr

Abstract

Analog and mixed-signal integrated circuit (AMS-IC) designs remain a critical yet challenging aspect within electronic design automation (EDA), primarily due to the inherent complexity, nonlinear behavior, and increasing variability associated with advanced semiconductor technologies. Traditional manual and intuition-driven methodologies for AMS-ICs design, which rely heavily on iterative simulation loops and extensive designer experience, face significant limitations concerning efficiency, scalability, and reproducibility. Recently, machine learning (ML) techniques have emerged as powerful tools to address these challenges, offering significant enhancements in modeling, abstraction, optimization, and automation capabilities for AMS-ICs. This review systematically examines recent advancements in ML-driven methodologies applied to analog circuit design, specifically focusing on modeling techniques such as Bayesian inference and neural network (NN)-based surrogate models, optimization and sizing strategies, specification-driven predictive design, and artificial intelligence (AI)-assisted design automation for layout generation. Through an extensive survey of the existing literature, we analyze the effectiveness, strengths, and limitations of various ML approaches, identifying key trends and gaps within the current research landscape. Finally, the paper outlines potential future research directions aimed at advancing ML integration in AMS-ICs design, emphasizing the need for improved explainability, data availability, methodological rigor, and end-to-end automation.

Keywords: analog design; mixed-signal design; machine learning; artificial intelligence; modeling; abstraction; optimization; sizing; specification-driven; assisted design



Academic Editor: Tee Hui Teo

Received: 14 August 2025

Revised: 31 August 2025

Accepted: 3 September 2025

Published: 5 September 2025

Citation: Liakos, K.G.; Plessas, F. Analog Design and Machine Learning: A Review. *Electronics* **2025**, *14*, 3541. <https://doi.org/10.3390/electronics14173541>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Analog and mixed-signal integrated circuits (AMS-ICs) represent essential components widely utilized across numerous applications, including wireless communications, medical devices, automotive systems, biosensors, and security electronics. Unlike digital circuits, AMS-ICs often involve intricate and nonlinear interactions, making their design particularly challenging [1]. With the continuous scaling of semiconductor technologies, the complexity and variability of these circuits have increased significantly, imposing additional difficulties on designers and electronic design automation (EDA) tools [1,2].

Traditionally, AMS-IC designs have relied heavily on expert intuition and manual tuning, where designers iteratively adjust transistor sizes, bias currents, and passive components to meet targeted performance specifications. This manual approach becomes increasingly inefficient and unreliable in advanced technology nodes due to severe process variations and complex physical phenomena [1]. Furthermore, the iterative design process contributes significantly to lengthy development cycles and increased time-to-market pressures [2].

In contrast, digital circuit design has greatly benefited from extensive automation and established EDA tools, significantly reducing design time and improving predictability and reproducibility. Unfortunately, the AMS-ICs design process has not experienced comparable levels of automation, partly due to the inherent complexity of analog design and the lack of effective modeling methods and tools capable of handling nonlinear circuit behavior [2,3].

In recent years, machine learning (ML) techniques have emerged as promising candidates to bridge this gap by providing computationally efficient methods for modeling, optimization, and design automation of AMS-ICs. ML methods, including artificial neural networks (ANNs), Bayesian models, reinforcement learning (RL), and Support Vector Machines (SVMs), have demonstrated notable success in capturing nonlinear relationships, managing high-dimensional variability, and enhancing the predictability of circuit performance [1,4]. Specifically, ML-driven approaches facilitate significant acceleration in analog circuit design by reducing the reliance on computationally expensive simulations and enabling the reuse of knowledge across design tasks and contexts [3,5].

This review paper systematically explores state-of-the-art ML techniques and methodologies applied to AMS-IC design. Section 2 presents an overview of ML fundamentals, introducing key paradigms such as supervised, semi-supervised, unsupervised, and RL learning tasks and describing the primary models associated with each category. Section 3 reviews the relevant literature and covers topics including circuit modeling and abstraction, optimization and sizing techniques, specification-driven design, and artificial intelligence (AI)-assisted design automation methods. Section 4 provides discussions and conclusions drawn from the surveyed works, while Section 5 highlights potential directions for future research in the field.

In this review, we provide a systematic survey of recent applications of machine learning techniques to analog and mixed-signal circuit design. The paper is organized as follows: Section 2 introduces the fundamental ML paradigms relevant to circuit design. Section 3 presents a structured review of the literature, categorized into modeling and abstraction, optimization and sizing, specification-driven design, and AI-assisted automation. Section 4 discusses emerging trends and key open challenges, while Section 5 outlines potential future research directions. Finally, Section 6 concludes the paper.

2. ML: An Overview

2.1. Terminology and Learning Paradigms

ML refers to computational algorithms and statistical techniques that enable computer systems to recognize patterns, learn from data, and make informed decisions without explicit programming [6,7]. Fundamental terms include features, representing variables describing the input data; labels, the output variables or targets; training sets for model building; validation sets for parameter tuning; and test sets for performance evaluation [7].

ML paradigms encompass supervised, semi-supervised, unsupervised, and RL, each characterized by unique methodologies and objectives [8,9]. These paradigms differ primarily in the availability of labeled data and the learning objective and play a central role in determining the choice of algorithms and models used in AMS-ICs design.

2.2. Supervised, Semi-Supervised, Unsupervised, and RL Tasks

ML tasks are generally categorized based on the availability of labeled data and the nature of the learning objective. Understanding these categories is essential for selecting appropriate models and methodologies in AMS-ICs design. The four primary learning paradigms are supervised, semi-supervised, unsupervised, and RL—each with distinct characteristics and applications.

2.2.1. Supervised Learning

Supervised learning is a fundamental paradigm in ML where a model is trained on a dataset comprising input–output pairs (x, y) . The objective is to learn a function, f , that maps input vectors $x \in R^n$ to target outputs y , by minimizing the discrepancy between the predicted and true values [10,11]. The relationship between the input and output is typically modeled as follows:

$$y = f(x; \theta) + \varepsilon \quad (1)$$

where $x \in R^n$ denotes the input feature vector, $y \in R$ is the corresponding output (dependent variable), $f(x; \theta)$ represents the predictive model parameterized by θ , and ε is an error term accounting for noise or model imperfections.

The model parameters θ are learned by minimizing a predefined loss function $\mathcal{L}$, such as the mean squared error (MSE) for regression tasks or the cross-entropy loss for classification problems. The optimization objective over a labeled dataset $(x_i, y_i)_{i=1}^N$ is formulated as follows:

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(f(x_i; \theta), y_i) \quad (2)$$

where N denotes the number of labeled training examples. The choice of the loss function $\mathcal{L}$ depends on the specific task and data characteristics [9,11].

2.2.2. Semi-Supervised Learning

Semi-supervised learning (SSL) is a learning paradigm that combines a small amount of labeled data with a large volume of unlabeled data to improve generalization [12,13]. By leveraging the structure inherent in the unlabeled data, SSL techniques can enhance model performance, particularly when labeled samples are scarce or costly to obtain [14].

Let the dataset be partitioned into a labeled subset D_L and an unlabeled subset D_U , such that

$$D = D_L \cup D_U \quad (3)$$

where $D_L = \{(x_i, y_i)\}_{i=1}^l$ denotes the labeled data and $D_U = \{x_j\}_{j=l+1}^{l+u}$ denotes the unlabeled data.

The goal is to learn a function $f(x; \theta)$ that generalizes well across both subsets. The typical training objective in semi-supervised learning is formulated as follows:

$$\min_{\theta} \left[\frac{1}{l} \sum_{i=1}^l \mathcal{L}_s(f(x_i; \theta), y_i) + \lambda \cdot \frac{1}{u} \sum_{j=1}^u \mathcal{L}_u(f(x_j; \theta)) \right] \quad (4)$$

where $\mathcal{L}_s$ is the supervised loss applied to labeled data, cross-entropy or MSE, $\mathcal{L}_u$ is an unsupervised loss that regularizes the model using unlabeled data (e.g., consistency loss, entropy minimization, or pseudo-labeling), and λ is a weighting parameter that controls the influence of the unlabeled loss [13,15].

This formulation allows the model to benefit from the abundance of unlabeled data by guiding its predictions to be consistent and confident, even when explicit supervision is limited. SSL has gained increasing traction in domains where labeling is expensive or domain expertise is required—making it particularly suitable for AMS-ICs design workflows involving large unlabeled simulation datasets [1].

2.2.3. Unsupervised Learning

Unsupervised learning refers to a class of ML techniques that aim to discover underlying patterns, structures, or latent representations from data without the use of labeled outputs [16]. This paradigm is particularly valuable when labeled data is scarce or unavail-

able and is commonly used in tasks such as clustering, dimensionality reduction, density estimation, and anomaly detection [17].

Formally, given a dataset of input samples $x \sim p(x)$, the objective is to transform or group these inputs into a latent representation z , such that

$$x \mapsto z \quad (5)$$

where $x \in R^n$, denotes the observed input data, $z \in R^m$, represents the learned latent variables, cluster indices, or lower-dimensional embeddings [11].

Two widely used unsupervised learning techniques include clustering, like K-Means, and dimensionality reduction, like Principal Component Analysis (PCA).

K-Means clustering partitions a dataset into K clusters by minimizing the within-cluster sum of squared distances. The optimization problem is given by the following:

$$\min_{\{C_k\}_{k=1}^K} \sum_{i=1}^N \min_k |x_i - C_k|^2 \quad (6)$$

where $x \in R^n$ is the i -th data point, $C_k \in R^n$ is the centroid of cluster k , and N is the total number of data points [17].

PCA is a linear dimensionality reduction technique that finds the directions of maximum variance in the data. It seeks to project the input data onto a lower-dimensional subspace while retaining the most informative components. The optimization objective is the following:

$$\max_W \text{Var}(W^T x) \text{ subject to } W^T W = I \quad (7)$$

where $W \in R^{n \times m}$ is the projection matrix ($m < n$), $\text{Var}(W^T x)$ is the variance of the projected data, and the constraint $W^T W = I$ ensures that the projection vectors are orthonormal [18].

2.2.4. RL

RL is a computational learning paradigm in which an agent interacts with an environment in a sequential decision-making process to learn an optimal policy. The agent receives feedback in the form of scalar rewards and aims to maximize the long-term cumulative reward [19,20]. This framework is typically modeled as a Markov Decision Process (MDP). An MDP is formally defined as a 5-tuple:

$$MDP = (S, A, P, R, \gamma) \quad (8)$$

where S is the set of all possible states, A is the set of all possible actions $P(s' | s, a)$ is the transition probability function defining the probability of transitioning to state s' given the current state s and action a , $R(s, a)$ is the reward function specifying the immediate reward received after taking action a in state s , and $\gamma \in [0, 1]$ is the discount factor, which balances the importance of immediate versus future rewards [20].

The goal of the agent is to learn a policy $\pi(a | s)$, which defines the probability of taking action a in state s , such that the expected return is maximized:

$$J(\pi) = E_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R_t \right] \quad (9)$$

To evaluate the quality of a policy, the state-value function $V^\pi(s)$ is defined as the expected return when starting from state s and following policy π thereafter:

$$V^\pi(s) = E_\pi \left[\sum_{t=0}^{\infty} \gamma^t R_t \mid s_0 = s \right] \quad (10)$$

The state-value function satisfies the Bellman equation, which expresses a recursive relationship:

$$V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a) [R(s, a) + \gamma V^\pi(s')] \quad (11)$$

This recursive formulation forms the basis for various dynamic programming and RL algorithms, such as value iteration, policy iteration, and Q-learning [21,22].

RL techniques are increasingly adopted in AMS-ICs design to enable closed-loop optimization, autonomous tuning, and exploration of design spaces where explicit modeling is complex or infeasible [5].

2.3. Learning Models

ML offers a diverse set of models suited to a range of data-driven tasks, from regression and classification to optimization and surrogate modeling. While many models exist—such as decision trees, ensemble methods, and probabilistic graphical models—certain families have proven especially effective in the context of AMS-IC design. This section focuses on three of the most widely adopted models in the AMS-IC literature: Bayesian models, SVMs, and ANNs. These models have demonstrated strong performance in tasks such as circuit modeling, performance prediction, optimization, and design automation. For completeness, foundational models like linear and logistic regression are also briefly discussed, as they serve as important baselines and are occasionally used in early-stage design exploration.

2.3.1. Bayesian Models

Bayesian models are probabilistic frameworks that incorporate prior knowledge and update beliefs using observed data. They are particularly well-suited for AMS-ICs design tasks where uncertainty quantification is critical, such as performance prediction under process variation or surrogate modeling [23,24].

Bayes' theorem provides the following basis:

$$P(\theta \mid D) = \frac{P(D \mid \theta)P(\theta)}{P(D)} \quad (12)$$

where $P(\theta \mid D)$ is the posterior probability of parameters θ given data D , $P(D \mid \theta)$ is the likelihood of the data under model parameters, $P(\theta)$ is the prior belief about parameters, and $P(D)$ presents the marginal likelihood normalization factor.

Bayesian models play a critical role in AMS-ICs design. They are employed in surrogate modeling to efficiently approximate circuit performance metrics, enabling rapid and cost-effective design space exploration. Bayesian optimization (BO) facilitates automated sizing and tuning of circuit parameters, offering a data-efficient strategy for navigating complex design landscapes [25].

Furthermore, Bayesian approaches to uncertainty modeling help quantify and mitigate the effects of post-layout variations, thereby improving the robustness and predictability of circuit performance. Figure 1 provides a general overview of the Bayesian inference process, showing how prior beliefs are updated with observed data to form the posterior distribution.

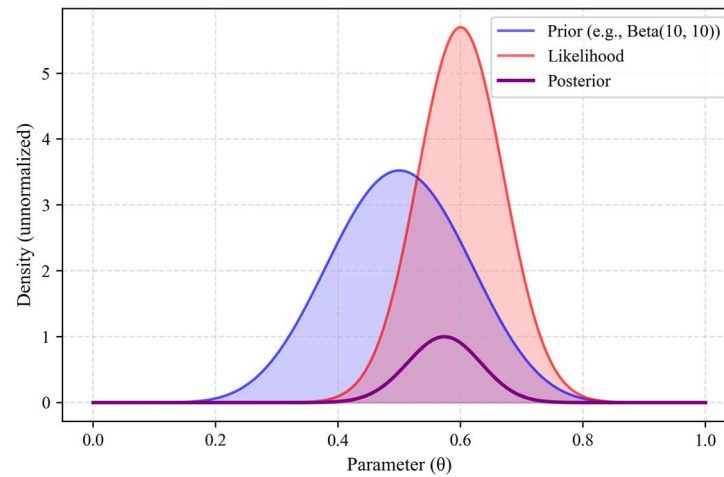


Figure 1. Visualization of Bayesian inference illustrating the relationship between the prior, likelihood, and posterior distributions.

2.3.2. ANNs

ANNs are a class of ML models inspired by the structure and functioning of the human brain. They consist of layers of interconnected nodes—commonly referred to as neurons—that can approximate complex nonlinear relationships between inputs and outputs [10]. Among various architectures, the feedforward neural network (FNN) is most widely adopted in AMS-IC design due to its simplicity and effectiveness in function approximation tasks. Figure 2 presents a general structure of a feedforward ANN with fully connected layers.

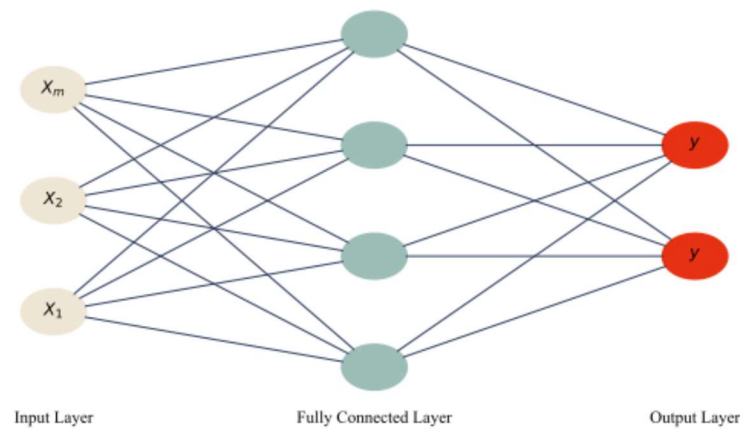


Figure 2. Structure of a feedforward ANN with fully connected layers.

The activation of a neuron in a feedforward layer is computed as follows:

$$a_j = \phi \left(\sum_{i=1}^n w_{ij} x_i + b_j \right) \quad (13)$$

where a_j denotes the output of neuron j , x_i represents the i -th input feature, w_{ij} is the weight connecting input i to neuron j , b_j is the bias term associated with neuron j , $\phi(\cdot)$ is a nonlinear activation function, such as the sigmoid, hyperbolic tangent (tanh), or rectified linear unit (ReLU) [7].

By stacking multiple layers of such neurons, ANNs are capable of learning highly expressive mappings between high-dimensional input and output spaces. The parameters w_{ij} and b_j are typically learned via backpropagation using gradient-based optimization.

In AMS-ICs design, ANNs are increasingly utilized due to their flexibility and ability to learn from large, noisy datasets. Key applications include specification-to-parameter mapping, where direct relationships are learned between high-level performance targets and transistor-level design parameters; layout generation and performance prediction, which involve estimating post-layout metrics from pre-layout features or assisting in layout synthesis; and post-layout regression modeling, which captures complex, nonlinear relationships between layout-induced parasitics and circuit-level performance. These capabilities make ANNs valuable tools for surrogate modeling, design space exploration, and layout-aware performance estimation within modern AMS-IC design flows.

2.3.3. SVMs

SVMs are supervised learning models widely used for both classification and regression tasks. They are particularly valued for their ability to construct robust decision boundaries by maximizing the margin between classes, thus enhancing generalization performance [26,27]. Figure 3 shows an SVM classification example with decision boundary and margin visualization.

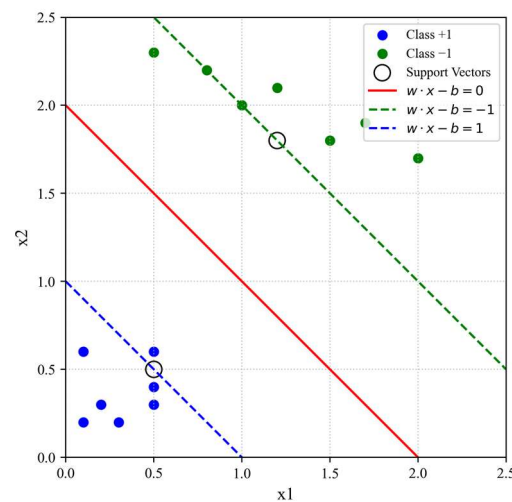


Figure 3. SVM classification with decision boundary and margin visualization.

In the case of linearly separable data, SVMs aim to find the hyperplane that best divides the data into distinct classes with the greatest possible margin. The decision function for a linear SVM is given by the following:

$$f(x) = w^T x + b \quad (14)$$

subject to the following constraint:

$$y_i (w^T x_i + b) \geq 1 \quad (15)$$

where $x_i \in R^n$ is the input feature vector, $y_i \in \{-1, +1\}$ is the class label associated with x_i , $w \in R^n$ is the weight vector defining the orientation of the hyperplane, and $b \in R$ is the bias term defining the offset [27].

For nonlinearly separable datasets, SVMs employ kernel functions to implicitly map the input data into a higher-dimensional feature space where linear separation becomes feasible. Common kernel choices include the Radial Basis Function (RBF), polynomial, and sigmoid kernels [26].

2.3.4. Linear and Logistic Regression

Linear and logistic regression are foundational statistical learning methods that serve as interpretable and computationally efficient tools in many ML tasks. In the context of AMS-ICs integrated circuit design, these models are commonly used for rapid performance estimation, classification, and as baseline models for benchmarking more complex approaches.

Linear Regression

Linear regression predicts a continuous outcome by modeling a linear relationship between input features and the target variable [28]. The predictive model is given by the following:

$$\hat{y} = \beta_0 + \sum_{i=1}^n \beta_i x_i \quad (16)$$

where $\hat{y}$ is the predicted response, x_i are the input features, β_0 is the intercept term, and β_i are the model coefficients learned from data (Figure 4).

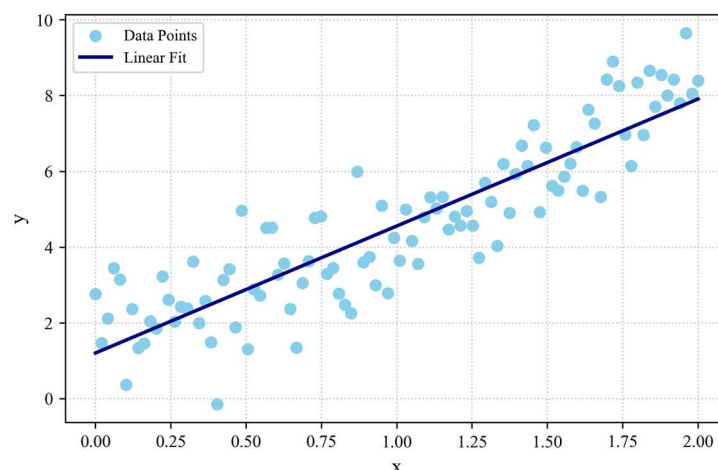


Figure 4. Linear regression visualization.

This method is most suitable when the underlying relationship between input and output is approximately linear and serves as a useful baseline in regression problems within AMS-ICs design.

Logistic Regression

Logistic regression is used for binary classification problems, modeling the probability that a given input belongs to a particular class [29]. The model uses the logistic (sigmoid) function to map real-valued inputs to a probability in the range [0, 1]:

$$P(y = 1 | x) = \frac{1}{1 + \exp(-(\beta_0 + \sum_{i=1}^n \beta_i x_i))} \quad (17)$$

where $y \in \{0, 1\}$ denotes the binary class label, and the parameters β_0, β_i are learned from data through maximum likelihood estimation [28,29] (Figure 5).

In AMS-ICs design, linear and logistic regression models remain valuable tools due to their simplicity, transparency, and low computational cost. They are particularly effective in performance region classification, helping to identify whether circuits meet acceptable performance thresholds. These models are also useful in early-stage design modeling, where they provide fast and interpretable rule-based approximations in scenarios with limited data or low design complexity. Additionally, they serve an important role in benchmark-

ing, offering a baseline for evaluating the performance gains achieved by more advanced models such as ANNs or ensemble methods, especially in contexts where interpretability and efficiency are critical.

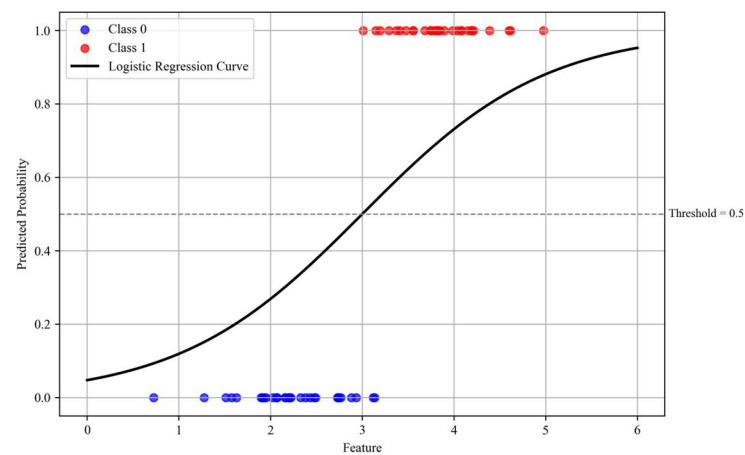


Figure 5. Logistic regression classification curve and data distribution.

2.3.5. Decision Trees

Decision trees (DT) are non-parametric ML models used for both regression and classification tasks. They operate by recursively partitioning the input space into distinct regions based on feature thresholds [30,31]. Each internal node in the tree corresponds to a decision rule on a feature, and each leaf node represents a prediction—either a class label (in classification) or a numeric value (in regression). Due to their interpretability and simplicity, DTs are particularly appealing for early-stage modeling and rule extraction in AMS-ICs design.

For regression tasks, the decision tree aims to minimize the variance of outputs within each region. The training objective can be defined as follows:

$$\min_{\text{splits}} \sum_{m=1}^M \sum_{x_i \in R_m} (y_i - \bar{y}_m)^2 \quad (18)$$

where R_m denotes the m -th terminal region (leaf node) defined by the tree structure, $\bar{y}_m$ is the mean of the target values y_i for all $x_i \in R_m$, and M is the total number of leaves (regions).

The model selects features and split points greedily, based on impurity reduction measures such as Gini impurity, entropy for classification, or variance reduction for regression [31] (Figure 6).

In AMS-ICs design, DTs are employed for a range of tasks due to their interpretability and adaptability. They are used in lightweight regression modeling to quickly approximate circuit behavior from limited datasets and in performance estimation under constraints by capturing simple, rule-based relationships between design variables and performance metrics [1,32]. Moreover, DTs serve as base learners in ensemble methods such as Random Forests and Gradient Boosted Trees, significantly enhancing prediction accuracy and robustness. Although individual DTs can be prone to overfitting, their integration into ensemble frameworks makes them a versatile and powerful component of ML workflows in AMS-ICs design.

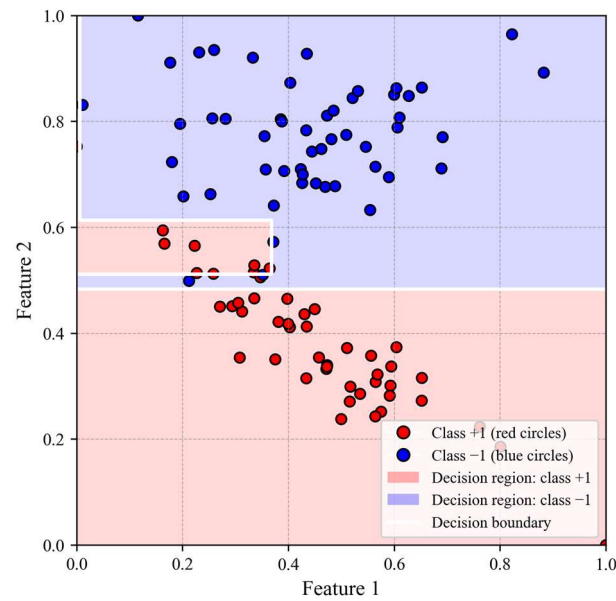


Figure 6. Decision tree classification boundaries in a two-dimensional feature space. Red circles denote class +1 samples and blue circles denote class −1 samples. The red and blue shaded areas indicate the classifier’s decision regions for classes +1 and −1, respectively; the white curve shows the decision boundary.

2.4. Evaluation Metrics

Evaluating the performance of ML models is essential for ensuring their reliability and suitability in design tasks. Depending on the learning objective—classification or regression—different metrics are used to assess model accuracy, robustness, and explanatory power [33].

2.4.1. Classification Metrics

For classification problems, performance is typically assessed using a set of metrics derived from the confusion matrix, which categorizes prediction outcomes into four groups: True Positives (*TP*), representing correctly predicted positive instances; True Negatives (*TN*), correctly predicted negative instances; False Positives (*FP*), negative instances incorrectly predicted as positive; and False Negatives (*FN*), positive instances incorrectly predicted as negative [34]. Based on these values, the core evaluation metrics are defined as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (19)$$

$$Precision = \frac{TP}{TP + FP} \quad (20)$$

$$Recall \text{ (Sensitivity)} = \frac{TP}{TP + FN} \quad (21)$$

$$F1\text{-score} = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (22)$$

Precision is particularly important in scenarios where false positives carry significant consequences—for example, incorrectly flagging valid circuit designs as invalid, which may lead to unnecessary design rework or discarded solutions. *Recall* becomes critical when false negatives are costly, such as missing potentially valid design candidates that meet performance requirements. The *F1-score* provides a balanced measure of *Precision* and *Recall*, making it especially valuable in cases of class imbalance, where relying solely on

accuracy may be misleading. This balance ensures that both types of misclassifications are considered when evaluating model performance.

2.4.2. Regression Metrics

For regression tasks, performance is typically evaluated using error-based metrics that quantify the difference between predicted and actual values [35]. MSE measures the average of the squares of the errors, emphasizing larger errors more due to squaring, which makes it sensitive to outliers. Mean absolute error (MAE) calculates the average of the absolute differences between predicted and actual values, offering a more interpretable measure of average error and being less affected by outliers [36]. The coefficient of determination (R^2) represents the proportion of variance in the dependent variable that can be explained by the independent variables; an R^2 value of 1 indicates perfect prediction, while a value of 0 means the model fails to capture any variability in the data [35,37]. These metrics together provide a comprehensive view of model accuracy and reliability in regression settings.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (23)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (24)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (25)$$

where y_i consists of the observed values, $\hat{y}_i$ represents the predicted values, $\bar{y}$ is the mean of observed values, and n is the number of observations.

3. Review for Analog Design Based on ML Methods

3.1. Literature Search Strategy and Studies Categorization

To ensure a systematic and reproducible review process, we conducted a structured literature search across several scientific databases, including IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Google Scholar. The search spanned publications from 2012 to early 2024, focusing on both journal and conference papers. We employed combinations of keywords such as “analog design”, “mixed-signal circuits”, “machine learning”, “deep learning”, and “artificial intelligence”.

Inclusion criteria required that the studies (i) presented a concrete application of ML techniques in analog or mixed-signal design, (ii) contained sufficient methodological detail, and (iii) were peer-reviewed. Exclusion criteria included works focusing solely on digital design, tutorial or survey articles without technical contributions, and papers with insufficient methodological clarity.

This methodology led to a final selection of 34 relevant works, which are systematically categorized and analyzed in the following subsections. Specifically, this review includes a total of 34 articles, comprising 29 conference papers and 5 journal articles (Figure 7), spanning the period from 2012 to 2024 (Figure 8). The studies are organized into four main thematic categories: Modeling and Abstraction of Analog Circuits, Optimization and Sizing Techniques, Specification-Driven Design, and AI-Assisted Design Automation.

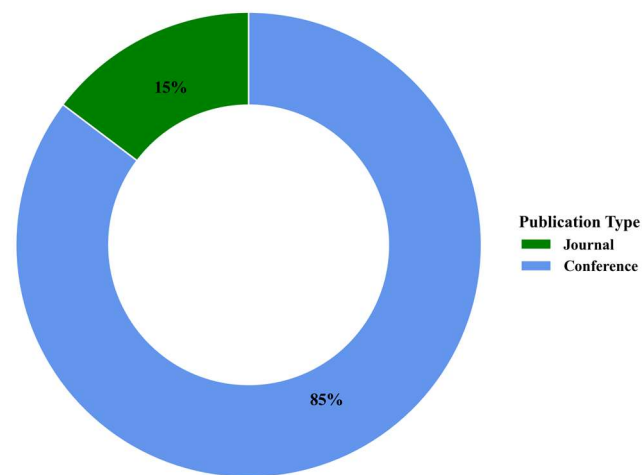


Figure 7. Categorization of studies by dissemination type.

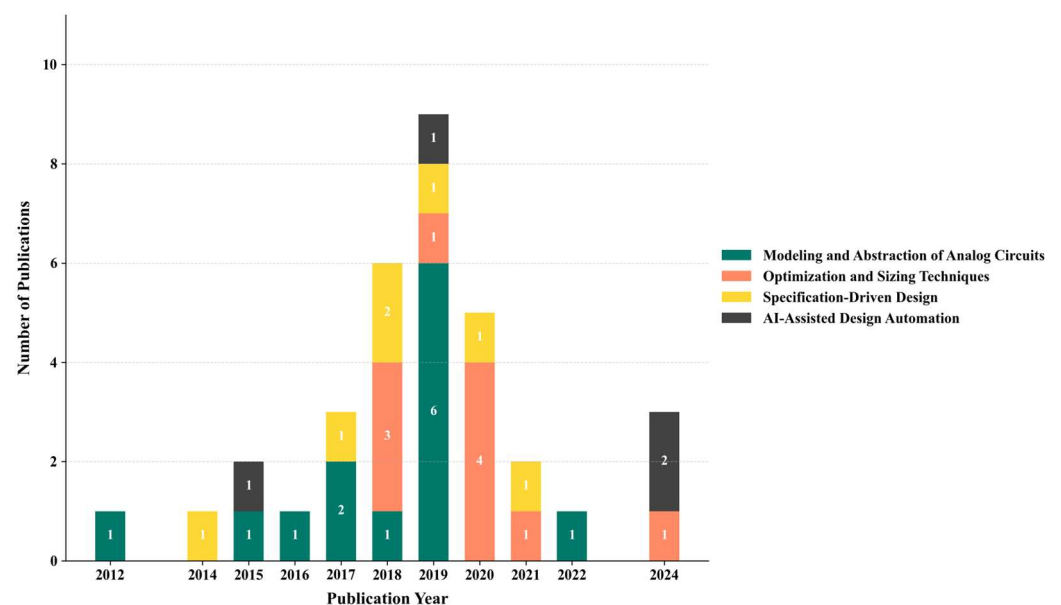


Figure 8. Annual distribution of publications by design methodology.

Specifically, the Modeling and Abstraction of Analog Circuits category includes thirteen studies, two journal articles, and eleven conference papers. Section 3.3 comprises ten articles, with three published in journals and seven in conferences. The Specification-Driven Design category includes seven conference papers, while the AI-Assisted Design Automation category consists of four conference publications (Figure 9).

3.2. Modeling and Abstraction of Analog Circuits

The complexity and nonlinearity inherent in analog ICs pose substantial challenges for efficient design, simulation, and verification. Traditional circuit modeling approaches, while accurate, are often computationally intensive and lack the flexibility to scale across diverse topologies or performance regimes. In recent years, ML techniques have emerged as powerful tools for abstracting analog circuit behavior, enabling faster simulation, improved performance prediction, and enhanced design reuse. Within this context, modeling methods based on ANNs, Gaussian processes, and Bayesian learning have demonstrated the ability to capture nonlinear circuit dynamics, learn from limited data, and generalize across design spaces. The works reviewed in this section employ these AI-driven models to create surrogate abstractions of analog blocks, ranging from op-amps to mixed-signal components, significantly accelerating simulation workflows and supporting early-stage

design decisions. This section outlines the main modeling strategies, abstraction levels, and application domains addressed in the recent literature.

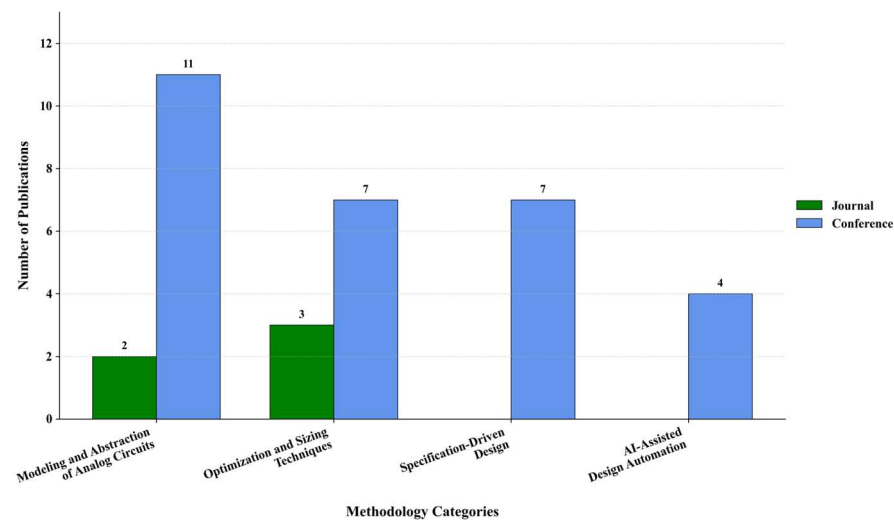


Figure 9. Distribution of journal and conference publications across AMS-ICs design methodology categories.

A Bayesian model fusion (BMF) technique for improving the efficiency of parametric yield estimation in AMS-ICs was introduced in [38]. The method utilizes simulation data from early design stages—such as schematic-level analysis—to construct informative priors for estimating complex post-layout performance distributions. A small number of late-stage simulations are then used in conjunction with maximum a posteriori (MAP) inference within a Bayesian framework to estimate yield. The approach was evaluated on ring oscillator (RO) and static random-access memory (SRAM) read path circuits, demonstrating up to a $3.75\times$ speed-up in simulation runtime over traditional kernel-based techniques, while maintaining high predictive accuracy. In [39], the authors introduced Co-Learning BMF (CL-BMF), an advanced performance modeling technique for AMS-ICs. CL-BMF combines coefficient side information (CSI) with performance side information (PSI) within a Bayesian inference framework represented as a graphical model. The key innovation lies in leveraging a low-complexity model—based on inexpensive performance metrics—to generate pseudo samples that facilitate the training of a high-complexity model using fewer costly simulations. Case studies on an RO and a 60 GHz low-noise amplifier (LNA) fabricated in 32 nm silicon-on-insulator (SOI) complementary metal oxide semiconductor (CMOS) demonstrate that CL-BMF achieves up to a $5\times$ reduction in modeling cost while preserving high accuracy. The same group [40] proposed BMF, a technique for efficient performance modeling of AMS-ICs. The method leverages early-stage simulation data as prior knowledge to support post-layout performance prediction, thereby reducing the number of late-stage simulations required. BMF models performance correlations between design stages using Bayesian inference, employing both zero-mean and nonzero-mean priors combined with MAP estimation. This allows the construction of accurate post-layout models, even in high-dimensional variation spaces. Experimental validation on an RO and an SRAM read path circuit fabricated in a 32 nm CMOS SOI process demonstrates up to a $9\times$ speed-up in runtime, along with improved accuracy over sparse regression methods. In [41], the authors proposed a compositional neural-network (CompNN) approach for modeling complex analog circuits as multiple input-multiple output (MIMO) systems. They utilized individual nonlinear autoregressive exogenous (NARX) networks to model specific input-output relationships like trimming, load jump, and line jump of a band-gap voltage reference (BGR) circuit. These outputs are then composed using a time delay neural network

(TDNN) to replicate the system behavior efficiently. The proposed model provides over $17\times$ speedup in transient simulation compared to transistor-level models while preserving accuracy. In [42], the authors introduced a hierarchical performance modeling methodology leveraging Bayesian Co-Learning (BCL). The approach decomposes the IC into hierarchical blocks and utilizes a combination of low-cost labeled and abundant unlabeled data to build accurate performance prediction models. Within this framework, they constructed a Bayesian model that integrates prior coefficient information, low-dimensional block-level models, and high-dimensional circuit-level models via semi-supervised learning. To reduce simulation cost, the method generates pseudo-labeled data from the low-level models. Experimental results on automatic delay compensation (ADC) delay-line and multi-stage amplifier circuits demonstrate notable runtime savings—up to $3.66\times$ —without compromising model accuracy. In [43], the authors proposed a self-adaptive multiple starting point (Smart-MSP) optimization approach for analog circuit synthesis. The approach integrates heuristic-biased starting point selection, sparse regression-based surrogate modeling, and probabilistic TABU (P-TABU) strategies to reduce simulation overhead. These mechanisms enable the optimizer to learn from past local searches and adaptively focus on high-potential regions within the design space. When compared to conventional optimization algorithms such as genetic algorithm (GA), simulated annealing (SA), particle swarm optimization (PSO), differential evolution (DE), and clustered simulated forced oscillation. Smart-MSP achieves a $2.6\times$ to $12.5\times$ speed-up while maintaining or improving solution quality. A parallelized implementation further enhances performance on large-scale synthesis tasks. Gao [44] proposed a Bayesian neural network (BNN)-based method for modeling performance trade-offs in analog circuit design. The authors propose using a single BNN, trained via automatic differential variational inference (ADVI), to simultaneously model multiple performance indicators of a circuit. This BNN is integrated into a BO framework that iteratively refines Pareto front approximations using a modified multi-objective evolutionary algorithm based on the decomposition (MOEA/D) evolutionary algorithm. Compared to traditional surrogate models such as Gaussian processes, the BNN approach more effectively captures correlations among points of interest (PoIs), thereby enhancing modeling efficiency and accuracy. Experimental results on charge pump and low-power amplifier circuits demonstrate a $2\times$ reduction in required simulations compared to prior methods. A hybrid analog IC synthesis method was proposed in [45], incorporating ANNs into a simulation-driven optimization flow guided by the SPEA2 evolutionary algorithm. Instead of discarding simulation data after each generation, the method reuses it to incrementally train ANNs as performance estimators. Once sufficiently trained, these networks partially replace the simulator, significantly reducing execution time. The approach was validated on a single-stage amplifier and a folded cascode operational transconductance amplifier (OTA), resulting in up to a 64.8% speed-up without degrading optimization performance. In [46], a Berkeley analog generator network (BagNet) was introduced as a layout-aware analog circuit optimization framework that integrates evolutionary algorithms with a deep neural network (DNN)-based oracle. The system utilizes the Berkeley analog generator (BAG) to create manufacturable layouts, while a DNN discriminator mimics simulation behavior, enabling selective evaluation of promising designs. By combining layout generation, simulation feedback, and ML-guided selection, BagNet achieved a reported $20\times$ to $300\times$ speed-up in post-layout optimization across a range of analog and radio frequency circuits (ARFCs), including operational amplifiers and optical receivers. A neural network (NN)-based Gaussian process regression (GPR) approach for BO was proposed in [47]. Unlike traditional GPR methods that rely on predefined kernels, this hybrid model uses an NN to learn feature embeddings that implicitly define the GP kernel. The resulting GP-NN model is paired with a weighted expected improvement (WEI) acquisition function and

enhanced through ensemble learning to improve uncertainty estimation. The approach demonstrated reduced simulation counts and computational cost in large design spaces, with successful application to operational amplifier and charge pump circuits. A sparse performance modeling approach based on spike-and-slab Bayesian learning was introduced in [48]. The method employs a hierarchical Bayesian mixture model to simultaneously identify relevant and irrelevant features by modeling each coefficient as a mixture of two zero-mean Gaussians: the “spike” for irrelevant features and the “slab” for important ones. Posterior inference is performed using Gibbs sampling. Results on a StrongARM latch comparator showed a 17% reduction in modeling error compared to least angle regression and ridge regression, while offering interpretable feature selection. In [49], the same research group introduces semi-supervised learning for efficient performance modeling (S2-PM) of analog and mixed-signal circuits. S2-PM exploits multiple representations of process variation—specifically, process variables and device-level variations—to incorporate unlabeled data into the training process without incurring additional simulation cost. The method co-trains two sparse regression models iteratively using both labeled and pseudo-labeled samples, where unlabeled data are selected based on a confidence metric derived from modeling error propagation. This co-learning strategy significantly reduces the required number of simulations while maintaining predictive accuracy. Finally, ref. [50] explored the use of ANNs for inverse modeling of analog amplifier performance. ANNs were trained using 4000 examples generated via the Python 3.6 simulation program with integrated circuit emphasis (Py-SPICE) simulations to learn the mapping from performance metrics—such as input impedance, gain, and power—back to circuit parameters (e.g., resistor values and transistor dimensions). Training strategies included adaptive learning rates, input/output normalization, and nonlinear activation functions such as tanh. The models demonstrated strong generalization in predicting impedance behavior and laid the foundation for advanced modeling pipelines in analog design.

A detailed summary of the reviewed works, including data size, modeling methods, and evaluation metrics, is presented in Table 1.

Table 1. Summary of studies in the Modeling and Abstraction of Analog Circuits category.

Cite	Data Size	Observed Features	AI Method	Evaluation	Results
[38]	RO: 4000 samples SRAM: 1000 samples	Power consumption, oscillation frequency, phase noise, and propagation delay	Semi-supervised BMF regressor	Prediction error	~0.2–2%
[39]	RO: 750 fundamental frequencies and 10 phase noise samples, LNA: 13 forward gain and 1 noise figure samples	Device-level process variations like threshold voltage, oscillation frequency, noise figure, and forward gain	Semi-supervised CL-BMF regressor	Relative error	~0.28–0.33 dB
[40]	Schematic: 3000 samples, Post-layout RO: 300 samples, Post-layout SRAM: 300 samples	Device-level variations and layout extracted parasitics	Supervised BMF regressor	Relative error	<2%

Table 1. Cont.

Cite	Data Size	Observed Features	AI Method	Evaluation	Results
[41]	Trimming: 695 samples, Load jumps: 433 samples, Line jumps: 501 samples	Trimming behavior and dynamic load/line transitions time-series signals	Supervised NARX and TDNN regressors	MSE, and R	Trimming: 7×10^{-5} , Load: 5.6×10^{-3} , Line: 1.4×10^{-3} , R > 0.96
[42]	Delay-line: 100 labeled and 300 unlabeled samples, Amplifier: 250 labeled and 450 unlabeled samples	Propagation delay and power consumption across hierarchical circuit blocks	Semi-supervised BCL regressor	Relative error	Delay-line: 0.13%, Amplifier: 0.55%
[43]	Simulated-oriented programmable amplifier: 3000–100,000 samples	Transistor sizing width and length, biasing parameters, voltage gain, and phase margin	Supervised G-OMP regressor	No explicit modeling error metric reported	Authors confirm equivalent or improved solution quality
[44]	Charge pump: 200 samples, Amplifier: 260 samples	Transistor width parameters and circuit-level performance metrics	Supervised BNN regressor	Hypervolume (HV) and Weighted gain (WG)	Charge pump: 15.04 HV and 0.19 WG, Amplifier: 10.72 HV and 102 WG
[45]	Over iterative design generations: 20,000 samples	Transistor sizing ratios width and length, bias currents, and power consumption	Supervised FNN regressor	No explicit modeling error metric reported	Authors confirm ANN outputs closely match SPICE simulation results
[46]	Optical receiver: 435 samples, DNN queries: 77,487 samples	Voltage gain, phase margin, diagram margin, and post-layout design specifications	Supervised DNN classifier	Sample compression efficiency	300×
[47]	Operational amplifier: 130 samples, Charge pump: 890 samples	Design specifications like voltage gain, phase margin, transistor dimensions (width and length), and passive component resistance and capacitance	Supervised Ensemble (BO GP NN) regressor	Figure of Merit (FOM)	Operational amplifier: FOM not reported Charge pump: 3.17 FOM
[48]	Labeled: 90 samples, Gibbs: 2800 samples	High-dimensional process parameters targeting power performance	Supervised ANN regressor	Relative error	2.39%

Table 1. Cont.

Cite	Data Size	Observed Features	AI Method	Evaluation	Results
[49]	Comparator: 50 orthogonal matching pursuit (OMP) samples and 70 S2-PM samples, Voltage-controlled oscillator: 50 OMP samples and 40 S2-PM samples, Unlabeled: 20 samples	Process variation parameters, including threshold voltage shifts	Semi-supervised Sparse regressor	Relative error	Comparator: OMP: 2.50%, S2-PM: 2.53% and Voltage- controlled oscillator: OMP: 1.55%, S2-PM: 1.6%
[50]	PySPICE: 4000 samples	Input impedance, voltage gain, power consumption, transistor dimensions (width and length), and passive component resistance and capacitance	Supervised FNN regressor	MSE	10^{-4} with ideal model, 10^{-2} with PySpice

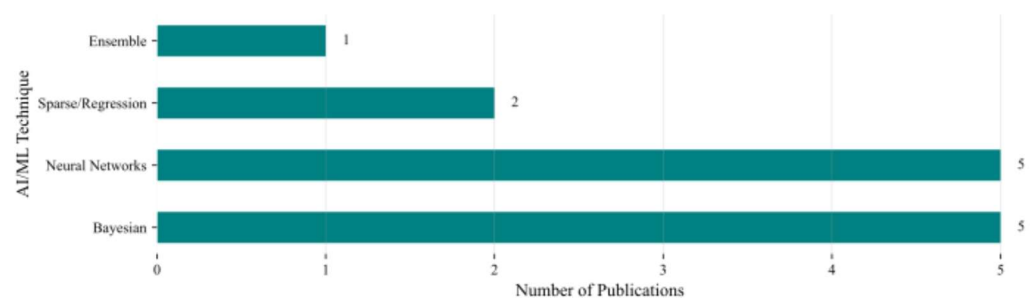
Overview of Modeling and Abstraction of Analog Circuits Techniques

Figure 10a provides a statistical overview of recent AI/ML-based modeling approaches applied to Modeling and Abstraction of Analog Circuits methods. As shown in Figure 10a, both Bayesian and NN-based methods are equally prominent, each appearing in five studies, reflecting the need for both probabilistic reasoning and nonlinear approximation capabilities. A smaller number of methods employ sparse regression (two studies) or ensemble learning (one study) indicating more specialized use cases.

Regarding the learning paradigm adopted, supervised learning is predominant, used in 69% of the studies, as shown in Figure 10b. Semi-supervised methods appear in 31%, highlighting growing interest in reducing simulation overhead by utilizing unlabeled or pseudo-labeled data.

As depicted in Figure 10c, the vast majority of works, 12 out of 13, formulate the modeling task as a regression problem, targeting accurate prediction of continuous circuit metrics such as delay, power, and gain. Only one study applies classification, suggesting limited application in this context.

Overall, these trends indicate a clear methodological preference toward Bayesian inference, NN architectures, and regression-based learning, driven by the need for accurate, data-efficient analog performance modeling.



(a)

Figure 10. Cont.

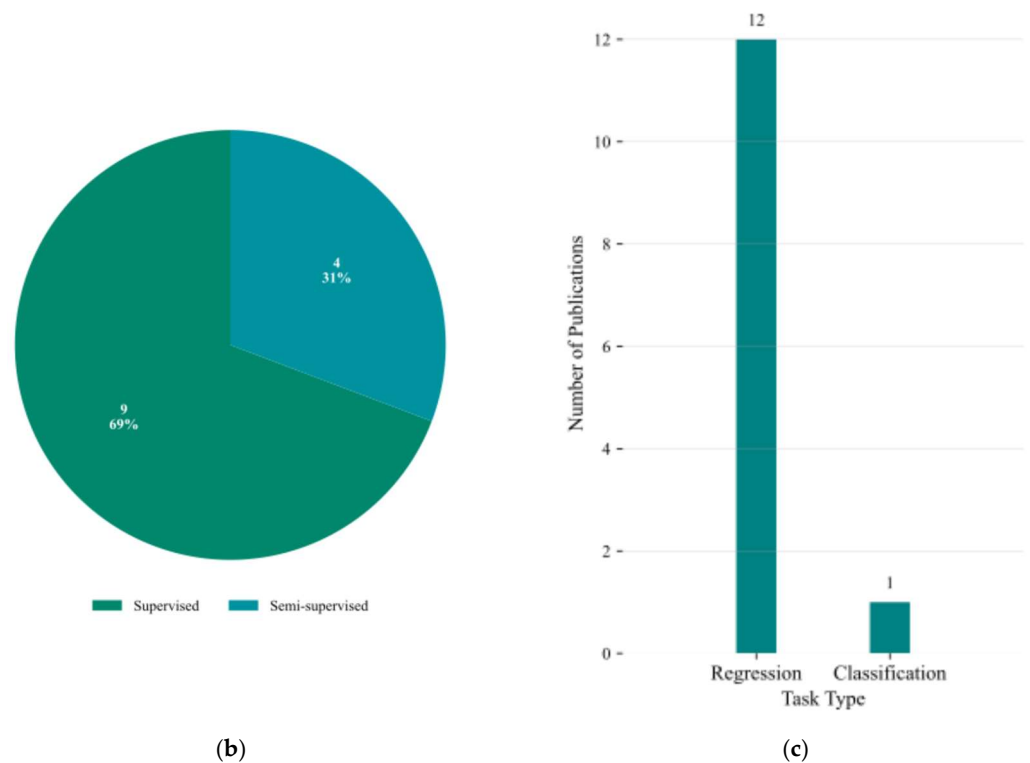


Figure 10. (a) Distribution of AI/ML techniques in Modeling and Abstraction of Analog Circuits category; (b) Types of learning approaches used in Modeling and Abstraction of Analog Circuits category; and (c) Prediction task in Modeling and Abstraction of Analog Circuits category.

3.3. Optimization and Sizing Techniques

Analog circuit sizing is a critical and resource-intensive phase in the design process, often requiring expert intuition and iterative tuning across a vast, nonlinear design space. Conventional methods—such as manual sizing, gradient descent, and evolutionary algorithms—struggle with scalability and typically demand thousands of expensive simulations. In recent years, ML techniques and RL have been increasingly applied to automate and accelerate the sizing process. These approaches aim to learn from data, adapt to diverse circuit topologies, and reduce the dependency on handcrafted heuristics. This section reviews key AI-driven methods for analog circuit optimization and sizing, highlighting innovations in Bayesian learning, NN-based modeling, policy learning, and hybrid search strategies. Collectively, these works demonstrate significant improvements in simulation efficiency, convergence speed, and design reusability.

A BO framework that integrates GPR with adaptive yield estimation to efficiently optimize both analog and SRAM circuits was introduced in [51]. The proposed method reduces the number and cost of yield evaluations by leveraging model uncertainty and employing a utility-based expected improvement acquisition function. To further improve efficiency, an adaptive yield estimator is incorporated, which relaxes precision requirements for low-yield candidates while focusing more accurate evaluations on promising designs. Experimental results across various analog building blocks and SRAM arrays demonstrate that this approach outperforms conventional techniques in simulation efficiency, achieving comparable or superior accuracy with significantly fewer evaluations. The WEI-BO framework was proposed in [52] as a BO method for automated analog circuit sizing. It integrates WEI as its acquisition strategy and employs GPs as surrogate models for both performance objectives and design constraints. The acquisition function guides search by incorporating the probability of constraint satisfaction, enabling more targeted exploration. WEI-BO is extended to support multi-objective optimization through Tchebysheff scalarization and smooth con-

straint relaxation. Validation on a range of analog and RFCs—including voltage-controlled oscillators, power amplifiers (PAs), charge pumps, inductors, and transformers—shows consistent outperformance over conventional methods, especially in high-dimensional and constraint-heavy design problems. A batch BO framework named MACE (Multi-objective Acquisition Ensemble) was introduced in [53]. MACE improves optimization by integrating multiple acquisition functions—lower confidence bound (LCB), expected improvement (EI), and probability of improvement (PI)—and selecting candidate batches from their Pareto front. This ensemble balances exploration and exploitation while supporting parallel evaluation. Applied to synthetic benchmarks and real analog circuits, including operational and class-E power amplifiers, MACE demonstrates faster convergence and superior solution quality compared to parallel BO approaches such as EI-LP, BLCB, qEI, and qKG. In [54], the authors presented a hierarchical synthesis framework for analog and RFC design that is augmented with ML to accelerate performance exploration. The methodology employs Bayesian linear regression for accurate device-level modeling and replaces computationally intensive SPICE simulations in the evaluation phase with SVMs. A dynamic sampling strategy is also introduced, which iteratively adjusts the search boundaries to concentrate sampling on the most promising regions of the design space. The framework is validated on folded-cascode operational amplifier and RF distributed amplifier (RFDA) designs in 65 nm CMOS technology, where it achieves up to a $9\times$ speed-up in runtime compared to traditional methods such as PAGE and exhaustive search, with no loss in solution quality. A fully parallelized analog optimization framework combining a GA with an ANN-based surrogate model was introduced in [55]. ANN models are trained in parallel to approximate SPICE simulations, guiding local minimum searches within promising regions of the design space. Coarse-grained global exploration is performed via GA, while fine-tuning is handled by ANN predictions. The framework, tested on a two-stage operational amplifier and a fifth-order active-RC complex bandpass filter, achieved a $4\times$ speed-up over GA-SPICE methods while maintaining comparable performance. In [56], the authors proposed a deep reinforcement learning (DRL) framework for automated analog circuit sizing. The sizing process is formulated as a policy gradient-based RL task, where the agent learns to optimize transistor dimensions through iterative interaction with a circuit simulation environment. To improve efficiency, a symbolic filter is introduced to pre-screen and eliminate non-promising configurations before invoking computationally expensive simulations. At the core of the framework is a policy gradient neural network (PGNN), which generates parameter updates based on state vectors representing current sizing configurations. Applied to a folded-cascode operational amplifier, the method demonstrates efficient convergence while significantly reducing the number of required SPICE simulations, highlighting its potential for practical design automation. In [57], the authors proposed learning to design circuits (L2DC). An RL-based framework for analog circuit design that operates without the need for large pre-collected training datasets. The method employs a deep deterministic policy gradient (DDPG) agent to iteratively explore the design space by observing circuit behavior through simulation and adjusting transistor-level parameters in response. A key feature of the framework is its ability to distinguish between hard constraints—such as gain and bandwidth—and soft optimization objectives like power and area. The agent learns a two-phase strategy: first ensuring that all constraints are met and then focusing on optimizing secondary targets. Evaluated on two transimpedance amplifier designs, L2DC achieves performance comparable to or better than that of expert human designers and BO methods, while demonstrating a $25\times$ to $250\times$ improvement in sample efficiency. AutoCkt, a deep RL framework for specification-driven analog synthesis, was proposed in [5]. Trained on sparse design samples, the agent generalizes across varying performance requirements and learns to map targets to circuit parameters through iterative simulation

feedback. Initially trained with schematic-level simulations, the model is transferred to post-layout optimization using layout-extracted/parasitic extraction (PEX) simulations within the BAG toolchain. AutoCkt outperforms GAs and hybrid ML-GA methods in convergence speed, generalization, and sample efficiency. In [58], the authors evaluate multiple ML strategies for automated analog circuit synthesis, with a focus on a novel global ANN model trained using data optimized through the transconductance-to-current ratio. The study compares three approaches: an RL-based method, L2DC, a supervised global ANN using forward propagation, and a GA-assisted local ANN trained via backward propagation. The proposed global ANN model achieves 93% accuracy on a two-stage operational amplifier designed in 65 nm CMOS, using significantly fewer simulations due to the compact and informative nature of gm/Id-based data. Although the GA-assisted local ANN is reported to be approximately four times faster when trained on traditional width/length sweep data, the gm/Id-driven approach reduces dataset size and simulation time substantially, achieving high performance with fewer than 7500 training samples. Lastly, ref. [59] introduced adaptive design optimization—large language model (ADO-LLM), a hybrid framework that integrates Gaussian process-based BO with LLMs. The framework enhances analog sizing by combining domain knowledge extraction from LLMs with efficient design-space exploration from BO. During optimization, the LLM uses in-context learning from a small number of high-quality designs to propose new candidates, while GP-BO handles convergence and uncertainty. Applied to a two-stage differential amplifier and a hysteresis comparator, ADO-LLM consistently outperforms both standalone GP-BO and LLM-based methods in figures of merit (FOMs) and constraint satisfaction.

A detailed summary of the reviewed works, including data size, modeling methods, and evaluation metrics, is presented in Table 2.

Table 2. Summary of studies in the Optimization and Sizing Techniques category.

Cite	Data Size	Observed Features	AI Method	Evaluation	Results
[51]	24 parameters and 50,000 Monte Carlo simulations used for yield estimation	Design parameters include sizing, yield, and process corner metrics	Supervised BO–GPR regressor	Failure rate	1%
[52]	Simulation budget of 100–1000 per circuit, including 20–40 initial design-of-experiment (DoE) samples	Design variables like transistor sizes, passive dimensions, biasing levels, and performance metrics like gain, efficiency, and area	Supervised BO–GPR regressor	No explicit modeling error metric reported	Demonstrated equal or superior Pareto quality with significantly reduced simulation cost
[53]	Operational amplifier: 500 samples, Power amplifier: 500 samples	Includes performance specifications such as gain, phase margin, unity gain frequency, power-added efficiency, and output power, along with design variables such as transistor dimensions, resistors, and capacitor values	Supervised BO–GPR regressor	No explicit modeling error metric reported	Demonstrated equal or superior Pareto front quality with 4–5× fewer simulations compared to non-dominated sorting genetic algorithm—II (NSGA-II) and genetic algorithm-based sizing of parameters for analog design (GASPAD)

Table 2. Cont.

Cite	Data Size	Observed Features	AI Method	Evaluation	Results
[54]	Genetic algorithm: 260 population samples and 16 sampling segments	Device sizing parameters such as transistor geometry and biasing levels, along with performance metrics including gain, power, and bandwidth	Supervised BLR regressor and SVM classifier	No explicit modeling error metric reported	Demonstrated comparable or improved performance metrics with up to 1245–1518% speed-up in circuit synthesis
[55]	Operational amplifier: 158 samples, Cascode band-pass filter circuit: 632 samples	Includes transistor widths, resistances, and capacitances, along with performance metrics such as gain, unity gain bandwidth, bandwidth, and phase margin	Supervised ANN regressor	Model error and R	<1% and 0.99
[56]	50,000 training tuples	Design variables such as transistor lengths, widths, and bias voltages, as well as performance attributes like DC gain, bandwidth, phase margin, and gain margin	Reinforcement PGNN regressor	No explicit prediction error metrics reported	Model performance evaluated via convergence of the reward function, defined as a weighted combination of normalized DC gain, bandwidth, phase margin, and gain margin
[57]	Three-stage transimpedance amplifier: 40,000 SPICE samples, Two-stage transimpedance amplifier: 50,000 SPICE samples	Circuit operating characteristics, including DC operating points and AC magnitude/phase responses, along with transistor-level parameters such as threshold voltage, transconductance, and saturation voltage	Reinforcement DDPG regressor	No explicit surrogate-model error metrics reported	Three-stage transimpedance amplifier: satisfied FOM score with a $250\times$ reduction. Two-stage transimpedance amplifier: reached 97.1% of expert-designed bandwidth with a $25\times$ improvement in sample efficiency
[5]	Transimpedance amplifier: 500 samples, Operational amplifier: 1000 samples, Passive-element filter: 40 samples	Design objectives include gain, bandwidth, phase margin, power, and area, evaluated against design-space parameters such as transistor widths/lengths, capacitor values, and resistor values	Reinforcement DNN-PPO regressor	No explicit surrogate-model error metric	Achieved $40\times$ higher sample efficiency than a genetic-algorithm baseline

Table 2. Cont.

Cite	Data Size	Observed Features	AI Method	Evaluation	Results
[58]	Width-over-length sweep: 65.534 samples, Transconductance-to-drain-current sweep: 7.361 samples, GA-guided local refinement: 632 samples	Design objectives include voltage gain, phase margin, unity gain frequency, power consumption, and slew rate	Supervised ANN and reinforcement DDPG regressors	Prediction score	Width-over-length sweep: 90%, Transconductance-to-drain-current sweep: 93%, GA-guided local refinement: 75.8% and DDPG 25–250× less sample-efficient than the supervised/GA pipeline
[59]	Five seed points plus twenty iterations, each evaluating one LLM-generated and four GP-BO candidates: 105 samples	Amplifier: 14 continuous design variables—all transistor widths/lengths plus compensation resistor R_z and capacitor C_c ; Comparator: 12 continuous design variables—width and length of six transistors	Supervised LLM-guided GP-BO regressor	No explicit surrogate-model error metric	Satisfies every design specification for both amplifier and comparator, achieves the top figure-of-merit among all methods, and reaches convergence with a 4× reduction in optimization iterations

Overview of Optimization and Sizing Techniques

Figure 11a presents the distribution of AI/ML techniques applied in optimization and sizing studies. NNs are the most prevalent, used in 6 of the 10 reviewed publications (60%), reflecting their capability to learn complex nonlinear relationships in high-dimensional design spaces. Gaussian process models are the second most common, appearing in three publications (30%), valued for their provision of calibrated uncertainty estimates in BO frameworks. Bayesian linear regression and SVMs are each reported once (10% each), indicating limited exploration of classical statistical learning in this domain.

The breakdown of learning approaches is shown in Figure 11b. Supervised learning is the dominant paradigm, employed in six publications (60%). Reinforcement learning is applied in three publications (30%), while hybrid approaches combining supervised and reinforcement learning are reported in one publication (10%). This distribution illustrates a methodological preference for data-driven modelling with labeled datasets, while still accommodating simulation-driven optimization through reinforcement learning in a smaller subset of works.

Figure 11c categorizes the studies by prediction task. The vast majority, 10 of the 11 tasks represented (91%), are formulated as regression problems, predicting continuous-valued circuit parameters or performance metrics. Only one instance of classification (9%) is reported, suggesting that discrete decision-making is rare in the context of optimization and sizing.

Taken together, these results indicate that optimization and sizing research is characterized by a strong emphasis on supervised NN regressors, with Gaussian process surrogates and RL occupying important but smaller roles, and classification tasks being largely absent.

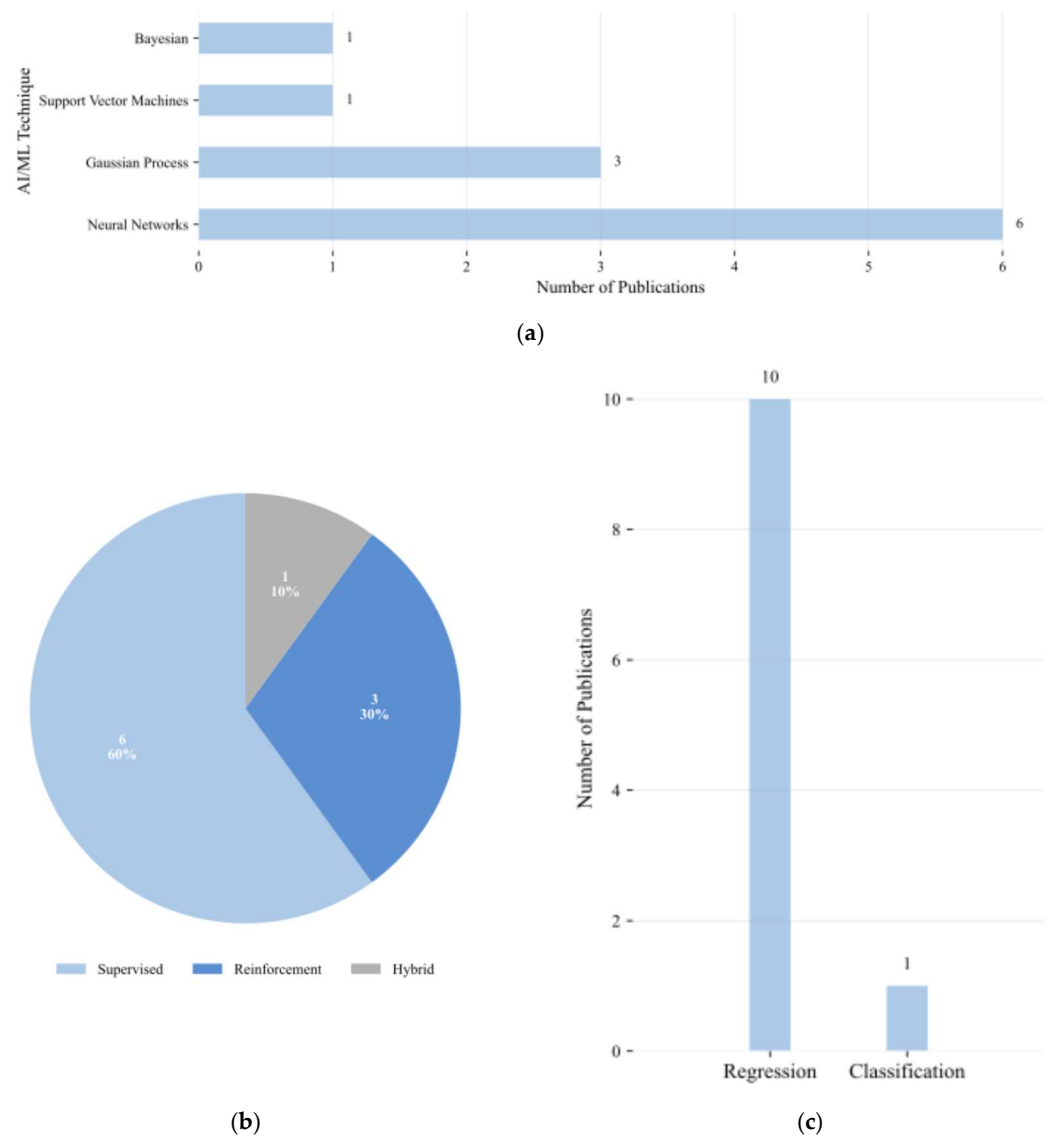


Figure 11. (a) Distribution of AI/ML techniques in the Optimization and Sizing Techniques category; (b) types of learning approaches used in the Optimization and Sizing Techniques category; and (c) prediction task in the Optimization and Sizing Techniques category.

3.4. Specification-Driven Design

Specification-driven design represents a shift from traditional iterative analog design toward direct inference of sizing parameters from performance specifications. Enabled by ML, these methods aim to bypass costly optimization loops by training models—typically ANNs—on simulated datasets to learn the mapping from specs to circuit elements. This approach significantly reduces design time and supports rapid reuse across different design contexts. Applications span a wide range of analog blocks, such as operational amplifiers and RFCs, achieving high accuracy with minimal simulations. In this section, we review key works that implement specification-driven methodologies, comparing their model types, data strategies, and performance outcomes.

A hybrid ML framework for automated synthesis of radio frequency low noise amplifier (RF-LNA) circuits was presented in [60], combining GAs with ANNs. The GA is used to optimize the ANN architecture (e.g., multilayer perceptron or radial basis function), input features, and hyperparameters. The resulting ANN predicts circuit component values—such as matching network elements and transistor dimensions—based on performance specifications. Trained on 235 RF-LNA designs simulated using a Tai-

wan Semiconductor Manufacturing Company (TSMC) 0.18 μm CMOS process, the model achieves over 99% prediction accuracy with an average response time of 62 ms per query. A regression-based approach for predicting component values in operational amplifiers was proposed in [61]. An FNN is trained on SPICE simulation data to map performance metrics such as DC gain, power consumption, and common-mode rejection ratio (CMRR) to corresponding sizing parameters. The trained model achieves approximately 90% accuracy and generates valid designs without additional simulations. Compared to conventional GA-based synthesis, this method reduces design time by up to $200\times$. In [62], the authors proposed a specification-driven approach for analog circuit sizing using ANNs trained to map performance specifications directly to sizing parameters. Departing from traditional simulation-intensive optimization flows, this method learns reusable design patterns from pre-optimized solutions, enabling fast inference without iterative evaluation. The study explores three ANN architectures applied to a voltage combiner-based amplifier, demonstrating the ability to generalize beyond training data and produce valid, high-quality sizing configurations with a significantly reduced number of simulations. In study [63], the authors evaluated two deep learning models—a recurrent neural network (RNN) and a deep feedforward neural network (DFNN)—for automatic sizing of analog operational amplifiers. Both models are trained on a synthetic dataset of 9000 SPICE-simulated circuits, each labeled with performance specifications including direct current (DC) gain, gain-bandwidth product (GBW), phase margin, CMRR, and power supply rejection ratio (PSRR). The networks learn to predict corresponding metal-oxide-semiconductor field-effect transistor (MOSFET) widths required to meet target specs. Results show that the DNN achieves higher prediction accuracy and stability, with a match rate of 95.6% compared to 92.6% for the RNN. In [64], the authors proposed a two-model framework aimed at enabling analog IC sizing reuse across varying design contexts, such as changes in supply voltage or load capacitance. The first model, termed context-independent performance estimator (CIPE), employs multivariate polynomial regression to estimate near-optimal performance trade-offs under new contextual conditions. The second model, circuit sizing predictor (CSP), is an ANN trained to predict the corresponding circuit sizing configuration that satisfies the estimated performance targets. This methodology facilitates fast, reusable design prediction without the need for repeated optimization. Applied to a folded cascode amplifier in a 130 nm CMOS process, the framework achieves results comparable to traditional optimization-based methods while reducing the required number of simulations by a factor of 400. An ANN-based method for automatic sizing of a two-stage operational amplifier was proposed in [65]. A single-hidden-layer feedforward ANN is trained on SPICE data to map six performance metrics—gain, phase margin, unity gain bandwidth, area, slew rate, and power consumption—to twelve design parameters, including MOSFET dimensions, bias currents, and compensation capacitance. Once trained, the model allows rapid prediction of valid configurations for unseen specifications, replacing simulation loops and reducing computational overhead. In the last study [66], the authors presented an NN-based method for automating the sizing of a two-stage CMOS operational amplifier. A dataset of 40,000 SPICE simulations was generated using linear technology (LT) LT-SPICE with randomized sizing parameters and performance evaluation scripts implemented in Python. The model is trained to predict device widths, lengths, capacitance, and bias current from a vector of performance specifications, including DC gain, phase margin, unity gain bandwidth, slew rate, and area. Once trained, the network provides rapid inference of valid sizing configurations without the need for iterative simulation.

A detailed summary of the reviewed works, including data size, modeling methods, and evaluation metrics, is presented in Table 3.

Table 3. Summary of studies in the Specification-Driven Design category.

Cite	Data Size	Observed Features	AI Method	Evaluation	Results
[60]	Low-noise amplifier: 235 samples	Bandwidth, 1-dB compression point, center frequency, third-order intercept point, noise figure, forward gain, transistor width/length ratio, source inductance, bias voltage	Supervised ANN regressor	Prediction accuracy	Matching networks: 99.22% accuracy; transistor geometries: 95.23% accuracy
[61]	20,000 samples after cleaning and normalization	Current and power consumption, direct current gain, phase margin, gain–bandwidth product, slew rate, total harmonic distortion, common and power rejection ratio, output and input voltage range, output resistance, input referred noise	Supervised FNN regressor	Average prediction accuracy	92.3%
[62]	Original: 16,600 samples, Augmented: 700,000 samples	Direct current gain, supply current consumption, gain bandwidth, phase margin	Supervised ANN regressor	MSE and MAE	0.0123 and 0.0750
[63]	Not precisely specified, 5.000–40.000 samples	Gain, bandwidth, power consumption	Supervised DNN regressor	Average match rate and RMSE	95.6% and ≈ 30
[64]	Pareto-optimal circuit sizing solutions: 700 samples	Load capacitance, circuit performance measures such as gain, unity gain frequency, power consumption, phase margin, widths and lengths	Supervised MP and ANN regressors	MAE	MP: 0.00807 and ANN: 0.00839
[65]	SPICE: 7.409 samples	Gain, phase margin, unity gain bandwidth, area, slew rate, power consumption, widths and lengths of selected transistors, bias current, and compensation capacitor	Supervised ANN regressor	MSE and R^2	$\approx 4.26 \times 10^{-10}$ and 94.29%
[66]	LT-SPICE: 40.000 samples	DC gain, phase margin, unity gain bandwidth, slew rate, area, compensation capacitor value and transistor widths, lengths, compensation capacitor, and bias current	Supervised FNN regressor	MAE	0.160

Overview of Specification-Driven Design Techniques

Figure 12a presents the distribution of AI/ML modelling approaches employed in recent specification-driven design studies. NNs were adopted exclusively across all the studies, indicating a complete dominance of this architecture within the category. This

prevalence reflects the well-documented capability of NNs to approximate complex, non-linear mappings in high-dimensional design spaces, a requirement intrinsic to translating circuit specifications into device-level parameters.

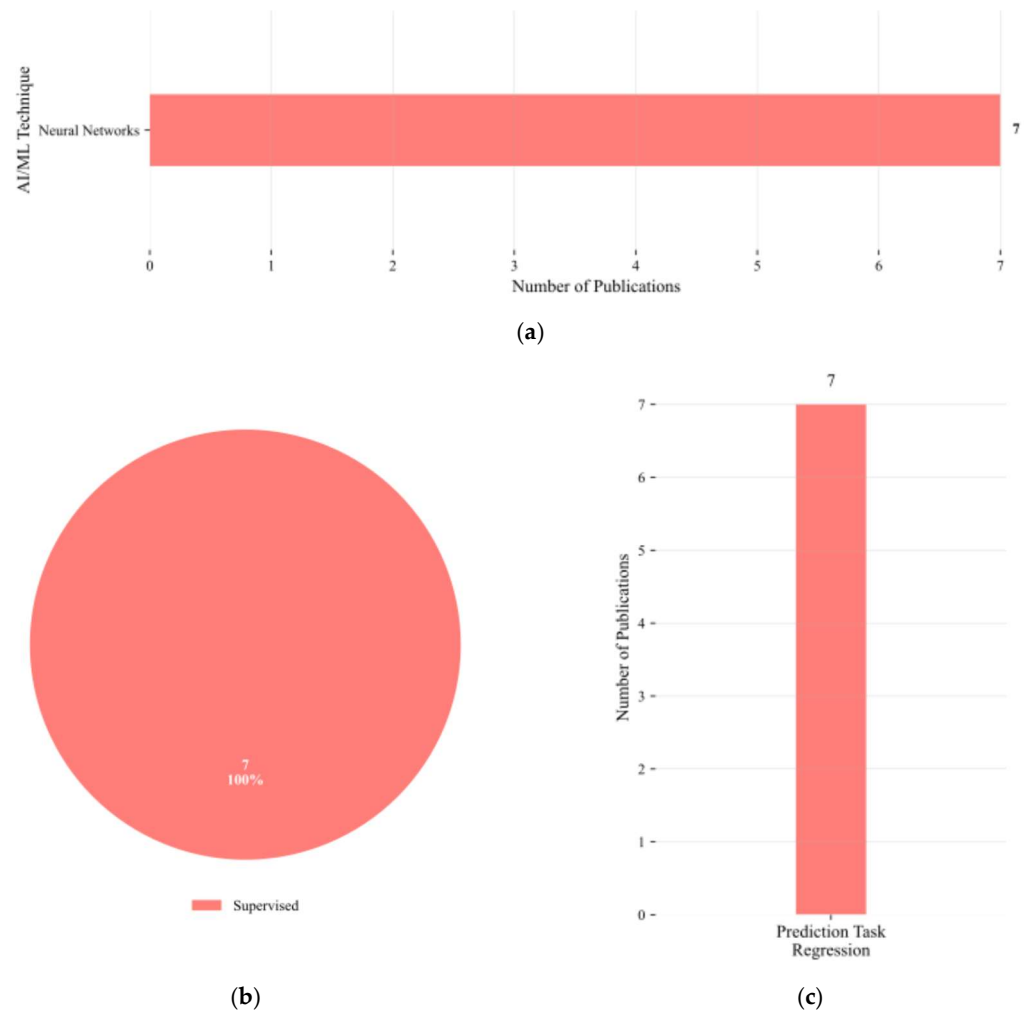


Figure 12. (a) Distribution of AI/ML techniques in the Specification-Driven Design category; (b) types of learning approaches used in the Specification-Driven Design category; and (c) prediction task in the Specification-Driven Design category.

The distribution of learning approaches is shown in Figure 12b. All reviewed works employed supervised learning, relying on pre-labeled datasets consisting of target specifications paired with corresponding design solutions.

Figure 12c reports the categorization by prediction task. All studies formulated the specification-driven design problem as a continuous regression task, mapping performance targets to continuous-valued design parameters. No instances of discrete classification tasks were observed, either as standalone objectives or in hybrid configurations.

3.5. AI-Assisted Design Automation

Beyond sizing and specification-driven flows, recent research has focused on leveraging AI to automate broader aspects of the analog design process. AI-assisted design automation encompasses methodologies that integrate learning-based techniques into layout generation, post-layout optimization, and adaptive design strategies under real-world constraints. These approaches typically incorporate RL, evolutionary algorithms, or ANNs to guide decision-making in tasks traditionally dominated by heuristic or rule-based flows. Such methods aim to accelerate late-stage design processes, reduce simulation over-

head, and enhance compatibility with industry-standard toolchains. This section presents key contributions in this domain, including RL-based floor planning, hybrid simulation-modeling pipelines, and layout-aware optimizations, highlighting their impact on design efficiency, scalability, and integration with commercial workflows.

An integrated seizure-detection IC incorporating ML techniques was developed in [67] to address analog and data-conversion non-idealities. The system includes an on-chip feature extraction engine and an embedded classifier capable of adapting to circuit-level variations. A probabilistic decision-boundary training mechanism ensures robustness against analog variability and quantization noise. The fabricated IC was validated in real-world seizure monitoring scenarios, demonstrating reliable detection performance under strict power and area constraints, and highlighting the potential of AI-assisted analog design in biomedical applications. In [68], the authors introduced a DNN work-enhanced evolutionary optimization framework for post-layout analog circuit sizing. The method integrates a DNN-based discriminator with a GA to guide the selection of promising design candidates, thereby reducing the number of required simulations. Implemented using the BAG toolchain, the framework is applied to the synthesis of an optical receiver front-end in a GlobalFoundries (GF) 14 nm process, requiring only 348 post-layout simulations. The approach achieves over 200× improvement in runtime and sample efficiency compared to conventional evolution-only methods, enabling efficient layout-aware optimization at advanced technology nodes. An AI-driven pipeline for analog layout generation combining RL and obstacle-avoiding Steiner routing was presented in [69]. The floorplanning task is formulated as an MDP and solved using a proximal policy optimization (PPO) agent, optionally augmented with simulated annealing. The global routing stage employs Steiner-tree-based algorithms to interface with the Analog AutoGENerator (ANAGEN) detailed router. The framework, trained on both synthetic and real designs, including operational transconductance amplifiers, demonstrated substantial improvements over metaheuristic baselines such as GA, SA, and PSO. Results show a reduction in layout generation time from 24 h to 21 min, with 14% area savings and 15% wirelength reduction. The system is also compatible with industrial flows, including integration into Infineon's design environment. Lastly, AICircuit [70] introduced a benchmarking infrastructure for AI-assisted analog and RF circuit design. It includes thousands of simulation-derived design points from seven analog/RFCs and two mm-wave systems, generated using Cadence. The platform enables standardized evaluation of ML-based inverse design methods—including MLPs, transformers, and support vector regression (SVR)—across diverse circuit architectures. While not a design methodology itself, AICircuit plays a vital role in assessing model performance and supporting the development of new approaches in analog design automation.

A detailed summary of the reviewed works, including data size, modeling methods, and evaluation metrics, is presented in Table 4.

Overview of AI-Assisted Design Automation Techniques

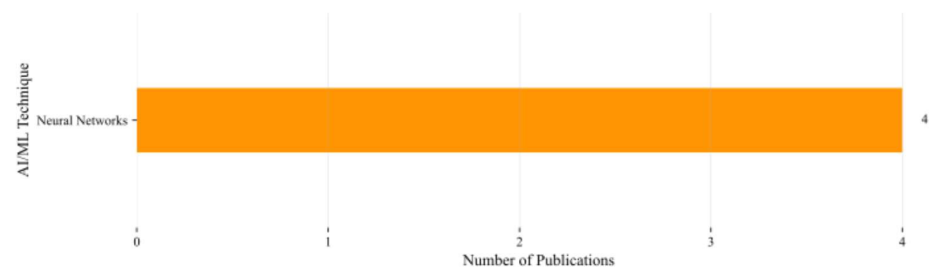
Figure 13a presents the distribution of AI/ML modelling approaches within the AI-assisted design automation category. NNs appear in all four reviewed publications, demonstrating their current dominance in this domain. This prevalence highlights their suitability for modelling complex relationships and decision processes inherent to automated design workflows.

Figure 13b shows the breakdown by learning approach. The studies are evenly split between supervised learning (2/4, 50%) and RL (2/4, 50%). This balance reflects the coexistence of methods that rely on pre-labeled datasets with those that learn through direct interaction with a simulation environment.

Table 4. Summary of studies in the AI-Assisted Design Automation category.

Cite	Data Size	Observed Features	AI Method	Evaluation	Results
[67]	Electroencephalogram segments: 4.098 samples	Frequency-domain energy in multiple bands and time-domain characteristics	Supervised ANN classifier	Accuracy, Recall	≈96% and ≈95%
[68]	Post-layout simulated designs: 338 samples	Design parameters of an optical link receiver front-end in 14 nm technology, including resistor dimensions, capacitor dimensions, and transistor properties such as number of fins and number of fingers; performance metrics such as gain, bandwidth, and other specification errors	Supervised DNN classifier	No explicit surrogate-model evaluation metric reported	Specification-compliant design in 338 simulations, 7.1 h, over 200× faster than baseline
[69]	Synthetic floorplans with 5–20 devices for training, an 11-device circuit for evaluation, and no explicit total sample count reported	Topological relationships between devices represented as sequence pairs; device parameters such as dimensions, shape configurations, symmetry, and alignment; performance objectives including occupied area and half-perimeter wirelength	Reinforcement DNN classifier	No explicit model-level evaluation metric reported	Reduced refinement effort by 95.8% and produced layouts 13.8% smaller and 14.9% shorter than manual designs
[70]	No explicit dataset size reported, multiple analog circuit topologies, including telescopic and folded cascode operational amplifiers	Gain, phase margin, power consumption, slew rate, transistor dimensions, bias currents, and compensation capacitance	Reinforcement DNN regressor	No explicit model-level evaluation metric reported	Fewer simulations, demonstrating higher sample efficiency in sizing multiple analog amplifier topologies

The distribution of prediction tasks is shown in Figure 13c. Three of the four publications (75%) address classification problems, such as distinguishing between design alternatives or pass/fail outcomes, while one publication (25%) formulates the problem as a regression task, predicting continuous-valued design parameters.



(a)

Figure 13. Cont.

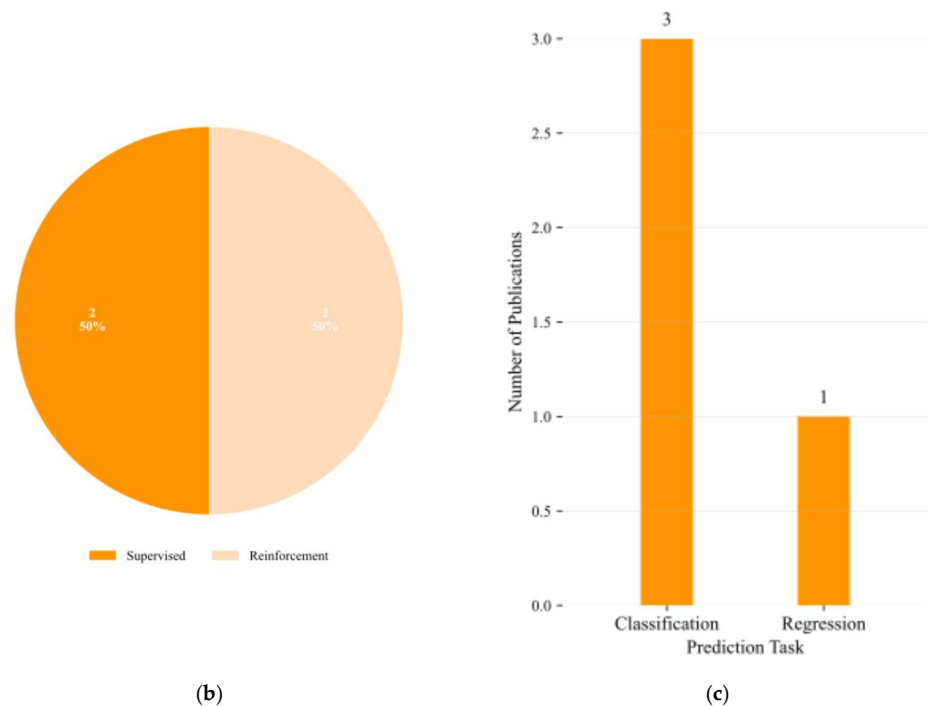


Figure 13. (a) Distribution of AI/ML techniques in the AI-Assisted Design Automation category; (b) types of learning approaches used in the AI-Assisted Design Automation category; and (c) prediction task in the AI-Assisted Design Automation category.

Taken together, these results indicate that AI-assisted design automation is characterized by exclusive reliance on NNs, with an equal split between supervised and reinforcement learning paradigms, and a predominance of classification tasks over regression formulations.

4. Discussion

In this review, we systematically examined recent contributions employing ML for front-end analog IC design automation. The collected works were analyzed based on their application category, ML algorithm type, learning paradigm, and task objective. From a total of 34 reviewed publications, four major application categories emerged (Figure 14):

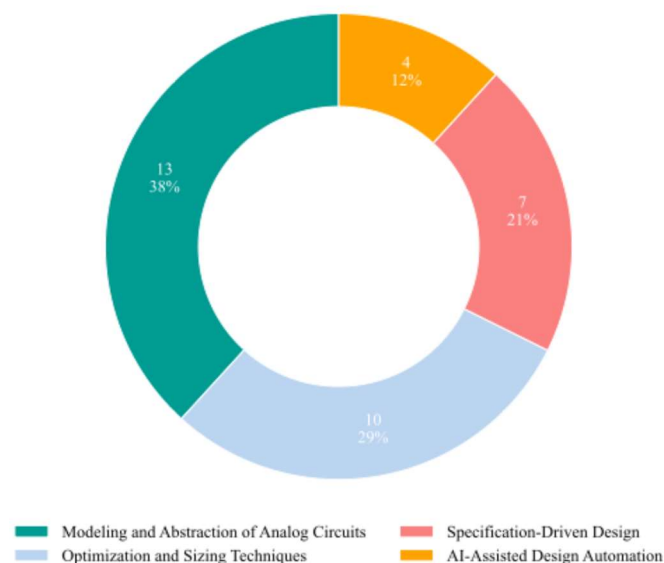


Figure 14. Pie chart presenting the approaches according to each category.

Modeling and Abstraction clearly dominates, representing 38% of the surveyed works (13 out of 34). This indicates that the community has invested significant effort in surrogate modeling and abstraction methods, largely motivated by the need to reduce the computational cost of post-layout simulations and to enable rapid design exploration. The second largest category is Optimization and Sizing Techniques, which accounts for 29% of the literature (10 studies). This share reflects the central role of parameter tuning and design-space exploration in analog IC design, where Bayesian optimization and reinforcement learning have shown particular promise. Specification-Driven Design contributes 21% of the reviewed papers (7 studies), underscoring growing interest in directly mapping high-level performance specifications to transistor-level implementations. While still a smaller fraction, these approaches demonstrate the potential to shorten design cycles and facilitate early-stage exploration. Finally, AI-Assisted Design Automation appears in only 12% of the works (4 studies), making it the least represented category. This scarcity highlights that, although initial attempts in ML-assisted placement, routing, and layout optimization exist, their integration into practical design flows remains in its infancy. Overall, the distribution suggests that the field has so far prioritized modeling and parameter optimization, with more limited attention to full automation and specification-driven design flows. This imbalance reflects the incremental path of ML adoption: researchers first address well-defined, simulation-intensive tasks before moving toward broader and more complex automation challenges.

The distribution of AI/ML techniques in Figure 15 reveals a clear methodological dominance of NNs, which appear in 22 out of 34 publications (64.7%). This overwhelming prevalence demonstrates the strong ability of neural architectures to capture nonlinear circuit behavior and high-dimensional design spaces. At the same time, it underscores the community's tendency to rely on black-box models despite their well-known limitations in interpretability and data efficiency. Bayesian approaches represent the second most common category, with six studies (17.6%). Their use is particularly prominent in modeling and abstraction tasks, where the ability to quantify uncertainty provides a significant advantage during early-stage design and verification. Gaussian process surrogates account for three papers (8.8%), reflecting their effectiveness in sample-efficient optimization but also highlighting scalability limitations in larger design spaces. Less frequently explored are regression methods (two works, 5.9%), SVMs (one work, 2.9%), and ensemble methods (one work, 2.9%). The marginal presence of these alternatives suggests that their potential in analog IC design remains underexplored, despite possible benefits in scenarios where data availability is limited or interpretability is prioritized. Overall, the figure indicates that the field is highly skewed towards NNs, with comparatively little systematic evaluation of alternative models. This imbalance suggests an opportunity for future work to critically assess algorithmic suitability across design tasks rather than relying predominantly on a single class of models.

Figure 16 illustrates how different ML techniques are distributed across the four methodological categories. The results show a strong methodological concentration toward NNs, while alternative approaches remain limited to specific niches. NNs dominate across all categories, appearing in 22 studies (64.7%) overall. Their distribution is as follows: four in Modeling and Abstraction (14.7%), six in Optimization and Sizing (17.6%), seven in Specification-Driven Design (20.6%), and four in AI-Assisted Design Automation (11.8%). This broad presence underscores their adaptability across diverse design tasks but also highlights the field's dependence on a single paradigm. Bayesian models account for six works (17.6%), with five concentrated in Modeling and Abstraction (14.7%) and one in Optimization and Sizing (2.9%). Their prevalence in modeling tasks reflects their strength in uncertainty quantification under limited or noisy datasets. Gaussian processes represent

three works (8.8%), all in Modeling and Abstraction. Regression methods appear in two studies (5.9%), also exclusively in Modeling and Abstraction. SVMs and Ensemble methods are each used in one study (2.9%), both within Optimization and Sizing. Taken together, over 80% of the literature relies on just two families of models: NNs and Bayesian methods, while alternative paradigms collectively account for less than 20%. This imbalance suggests that methodological choices are often driven by convention rather than systematic benchmarking across tasks. Expanding the methodological toolkit—particularly in under-explored areas such as AI-assisted automation and cross-abstraction integration—could reveal cases where non-neural approaches offer superior data efficiency or interpretability.

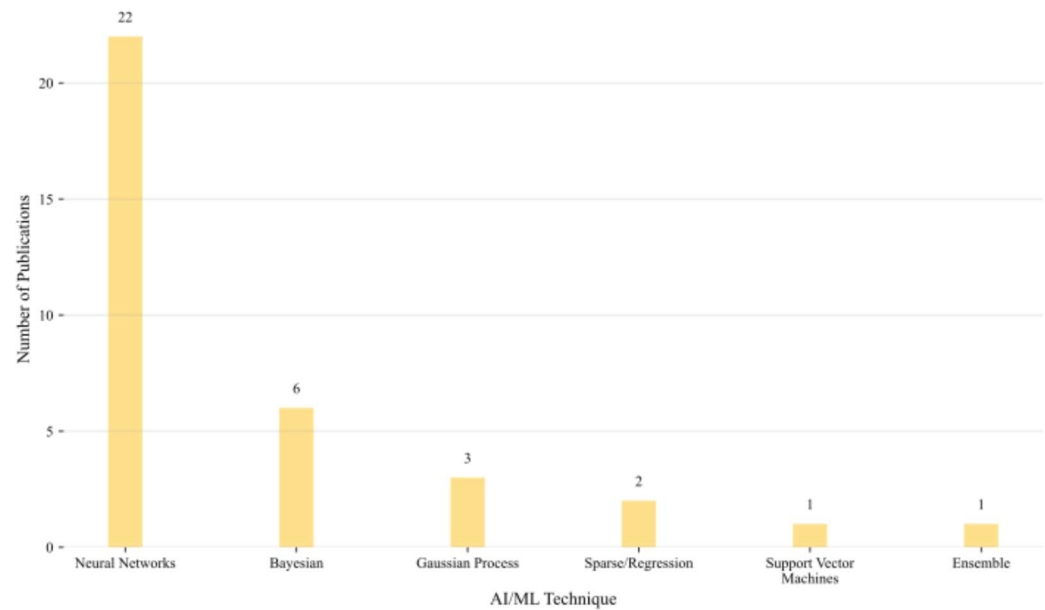


Figure 15. Distribution of AI/ML techniques.

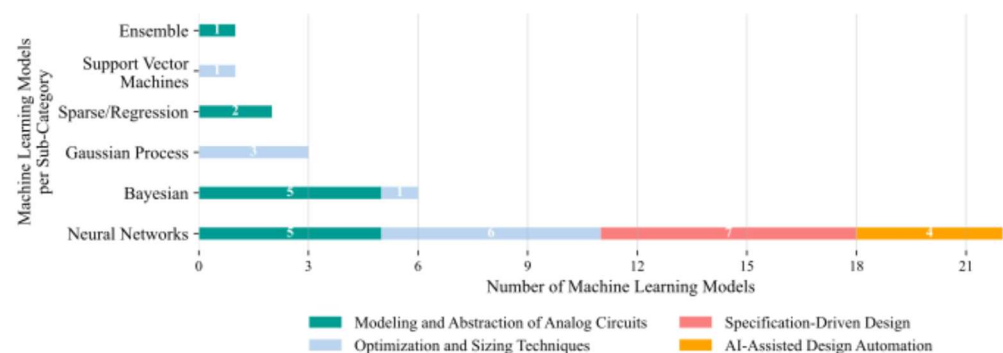


Figure 16. Distribution of ML techniques across methodological categories.

In terms of learning paradigm, supervised learning overwhelmingly dominates, appearing in 24 out of 34 studies (70.6%). Its distribution spans nine in Modeling and Abstraction (26.5%), six in Optimization and Sizing (17.6%), seven in Specification-Driven Design (20.6%), and two in AI-Assisted Design Automation (5.9%). This concentration reflects the field's reliance on labeled datasets and deterministic training strategies, which, while effective, may limit adaptability in data-scarce or exploratory design scenarios. RL accounts for five works (14.7%), primarily within Optimization and Sizing (8.8%) and AI-Assisted Design Automation (5.9%). These studies leverage simulator-based feedback to explore design spaces without pre-labeled data, showcasing RL's suitability for iterative optimization and search-intensive tasks. Semi-supervised learning appears in four works

(11.8%), all within Modeling and Abstraction, indicating early attempts to exploit partially labeled datasets for surrogate modeling but still far from mainstream adoption. Hybrid strategies are rare, with only one study (2.9%) combining supervised and RL approaches to balance predictive accuracy with exploratory efficiency (Figure 17). Overall, more than 70% of the literature remains tied to supervised learning, while RL, semi-supervised, and hybrid paradigms collectively represent less than one-third of contributions. This imbalance highlights the need for greater methodological diversity, especially in tasks such as variability-aware modeling, cross-abstraction integration, and real-time layout synthesis, where alternative learning paradigms may offer distinct advantages.

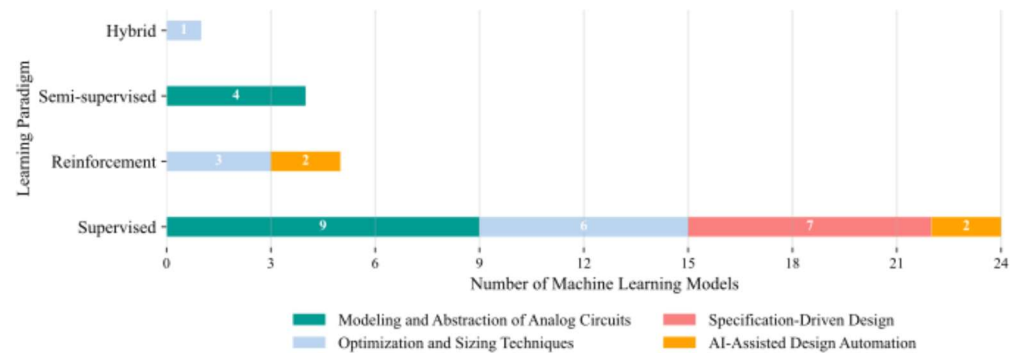


Figure 17. Distribution of learning types across methodological categories.

In terms of ML task type, regression clearly dominates, appearing in 30 out of 34 reviewed studies (88.2%). Its distribution includes 12 in Modeling and Abstraction (35.3%), 10 in Optimization and Sizing (29.4%), 7 in Specification-Driven Design (20.6%), and 1 in AI-Assisted Design Automation (2.9%). This overwhelming prevalence reflects the central role of regression-based approaches in analog design, where predicting continuous performance metrics such as gain, bandwidth, power consumption, or delay is critical. The reliance on regression also highlights the community's focus on quantitative performance prediction over discrete decision-making tasks. Classification methods are comparatively rare, used in only four studies (11.8%). These include one in Modeling and Abstraction (2.9%), one in Optimization and Sizing (2.9%), and two in AI-Assisted Design Automation (5.9%) (Figure 18). Their limited presence indicates that categorical decision tasks—such as topology selection, pass/fail verification, or layout constraint classification—remain underexplored in ML-driven analog design. Overall, nearly nine out of ten studies employ regression models, underscoring a methodological bias toward continuous prediction tasks. While effective for performance estimation and sizing, this imbalance suggests that the potential of classification approaches is not fully leveraged. Expanding research in classification—particularly for topology recommendation, design space partitioning, and constraint satisfaction—could complement regression and support more holistic automation workflows.

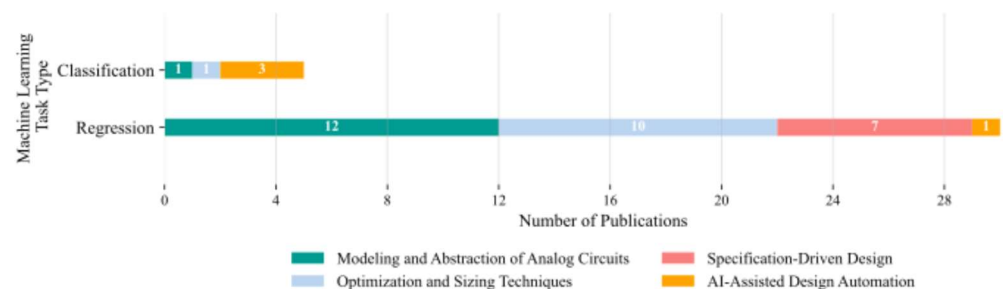


Figure 18. Contribution of methodological categories per ML task type.

Beyond methodological trends, an important aspect concerns the industrial relevance of the surveyed approaches. While these methods demonstrate significant academic advances, their practical adoption in analog design flows requires attention to several key factors. Tool compatibility is essential, as ML frameworks must integrate seamlessly with established EDA environments such as Cadence, Synopsys, or Siemens. Runtime efficiency also remains a critical concern; surrogate modeling and Bayesian optimization have shown particular promise in reducing simulation overhead, aligning ML methodologies with stringent time-to-market constraints. Designer trust is another challenge, underscoring the need for explainable AI (XAI) mechanisms that provide interpretable insights into model behavior. Finally, several recent works, including AutoCkt, L2DC, BagNet, and ADO-LLM, illustrate how academic innovations can transition into industrial contexts, achieving measurable improvements in convergence speed, design quality, and scalability. Collectively, these considerations emphasize that beyond accuracy, successful industrial deployment of ML in analog design will depend on interpretability, efficiency, and seamless integration into existing toolchains.

Building upon these insights, we now highlight several promising research directions that can address current limitations and further advance ML-driven AMS-ICs design.

5. Future Work

Despite the rapid adoption of ML techniques in analog IC design, several research avenues remain underexplored. Addressing these gaps will be critical for advancing the maturity of the field and enabling industrial-grade automation workflows:

(1) Methodological diversification. The current landscape is dominated by ANNs, which account for the majority of reported approaches. While powerful, ANNs face limitations in terms of explainability, data efficiency, and robustness across technologies. Future research should systematically evaluate alternative paradigms, including Bayesian learning, SVMs, ensemble methods, and graph neural networks (GNNs). Such diversification could reveal contexts where these methods outperform black-box neural models, particularly under data scarcity or stringent interpretability requirements.

(2) Explainability and trust. The integration of XAI frameworks into analog design workflows remains largely absent. Future studies should prioritize interpretable models that allow designers to identify which features and circuit parameters drive predictions. Transparent ML systems will not only improve debugging and model validation but also foster greater trust in automated design recommendations, thereby accelerating industrial adoption.

(3) Balanced exploration of learning paradigms. Supervised learning currently dominates the literature, whereas RL and semi-supervised approaches are comparatively underutilized. These paradigms offer natural advantages in exploration-heavy or low-label scenarios, such as design space traversal, on-the-fly layout synthesis, and adaptive verification. Hybrid strategies that combine predictive power from supervised models with the exploratory capabilities of RL also merit deeper investigation.

(4) Benchmarking and reproducibility. A major barrier to progress is the absence of open and standardized datasets. Establishing public benchmarks that include diverse circuit topologies, variation-aware conditions, and consistent performance metrics would significantly strengthen reproducibility and comparability across studies. Synthetic data generation, transfer learning pipelines, and industry-academia collaborations could further enrich the dataset ecosystem, enabling more robust and generalizable ML models.

(5) Leveraging LLMs. The emergence of foundation models opens an exciting new frontier: domain-specific LLM-driven conversational agents for analog design. Such systems could support specification clarification, simulation parameter tuning, design

troubleshooting, and even interactive guidance through the design flow. When coupled with circuit simulators and EDA environments, LLMs may evolve into intelligent assistants that enhance productivity for experienced practitioners and reduce the learning curve for new designers.

(6) End-to-end design integration. Finally, a key challenge is to move beyond isolated tasks and achieve holistic automation across the full AMS-ICs design flow. Future research should explore hierarchical learning architectures, transfer learning between design stages, and closed-loop optimization strategies that integrate feedback from simulators into iterative ML training. Achieving this vision would enable seamless, cross-abstraction automation, spanning from early modeling and sizing to layout generation and verification.

In summary, future research should aim not only to improve predictive accuracy but also to establish transparent, reproducible, and end-to-end design frameworks. By addressing methodological, data-centric, and integration challenges, the community can bridge the gap between academic innovation and industrial adoption, ultimately paving the way for next-generation ML-driven EDA tools capable of tackling the growing complexity of analog and mixed-signal circuits.

6. Conclusions

In summary, this paper presented a comprehensive review of recent applications of machine learning in analog and mixed-signal circuit design, covering 34 peer-reviewed contributions published between 2012 and 2024. The surveyed works were organized into four major categories: modeling and abstraction, optimization and sizing, specification-driven design, and AI-assisted automation. This taxonomy allowed us to identify methodological trends, emerging paradigms, and persistent gaps across the literature.

A key observation is the dominance of neural network-based approaches, which account for nearly two-thirds of the reviewed studies. While their success demonstrates the strong representational capacity of deep architectures for capturing nonlinear analog circuit behavior, it also exposes a lack of algorithmic diversity. Bayesian models and Gaussian processes remain valuable, particularly in settings where uncertainty quantification is critical, but their adoption is relatively limited. Reinforcement learning has shown promising results in sizing and layout optimization, yet its utilization is still modest compared to supervised learning, reflecting the field's dependence on labeled datasets.

From an application perspective, specification-driven design has benefitted the most from supervised learning, enabling direct mapping from performance specifications to transistor sizing with notable improvements in design turnaround time. Optimization and sizing tasks have leveraged Bayesian optimization and deep reinforcement learning to significantly reduce the number of costly simulations required, while modeling and abstraction methods have focused on surrogate modeling strategies to accelerate post-layout verification. AI-assisted design automation remains the least explored but highly impactful direction, where early demonstrations of ML-driven floorplanning, layout generation, and routing point toward future industrial adoption.

Despite these achievements, several limitations remain. Data scarcity and the absence of open benchmark datasets hinder reproducibility and fair comparison across methods. Explainability is rarely addressed, leaving designers with limited insight into how ML models arrive at predictions. Cross-abstraction generalization—from schematic to layout—remains an open challenge, preventing seamless end-to-end automation flows. Finally, evaluation rigor is inconsistent, with limited testing across technologies, process corners, or circuit topologies.

Taken together, these findings highlight that while ML has already demonstrated substantial benefits for AMS-ICs design, the field is still at an early stage of maturity. Ad-

addressing current gaps through methodological diversification, transparent and explainable models, robust evaluation frameworks, and open datasets will be critical to unlock the full potential of ML in this domain. We believe that continued collaboration between academia and industry will be essential for translating these advances into practical EDA tools capable of meeting the growing complexity and performance demands of next-generation analog and mixed-signal systems.

Author Contributions: Conceptualization, K.G.L. and F.P.; methodology K.G.L.; software, F.P.; validation, K.G.L. and F.P.; formal analysis, K.G.L.; investigation, F.P.; resources, K.G.L.; data curation, F.P.; writing—original draft preparation, K.G.L.; writing—review and editing, K.G.L. and F.P.; visualization, F.P.; supervision, K.G.L.; project administration, K.G.L.; funding acquisition, F.P. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Mina, R.; Jabbour, C.; Sakr, G.E. A Review of Machine Learning Techniques in Analog Integrated Circuit Design Automation. *Electronics* **2022**, *11*, 435. [CrossRef]
2. Afacan, E.; Lourenço, N.; Martins, R.; Dündar, G. Review: Machine learning techniques in analog/RF integrated circuit design, synthesis, layout, and test. *Integration* **2021**, *77*, 113–130. [CrossRef]
3. Cao, W.; Benosman, M.; Zhang, X.; Ma, R. Domain knowledge-infused deep learning for automated analog/radio-frequency circuit parameter optimization. In Proceedings of the 59th ACM/IEEE Design Automation Conference, San Francisco, CA, USA, 10–14 July 2022. [CrossRef]
4. Budak, A.F.; Bhansali, P.; Liu, B.; Sun, N.; Pan, D.Z.; Kashyap, C.V. DNN-Opt: An RL inspired optimization for analog circuit sizing using deep neural networks. In Proceedings of the 2021 58th ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, USA, 5–9 December 2021. [CrossRef]
5. Settaluri, K.; Haj-Ali, A.; Huang, Q.; Hakhamaneshi, K.; Nikolic, B. AutoCkt: Deep reinforcement learning of analog circuit designs. In Proceedings of the 2020 Design, Automation and Test in Europe Conference and Exhibition, (DATE), Grenoble, France, 9–13 March 2020. [CrossRef]
6. Sarker, I.H. Machine Learning: Algorithms, Real-World Applications and Research Directions. *SN Comput. Sci.* **2021**, *2*, 160. [CrossRef]
7. LeCun, Y.; Bengio, Y.; Hinton, G. Deep learning. *Nature* **2015**, *521*, 436–444. [CrossRef]
8. Jordan, M.I.; Mitchell, T.M. Machine learning: Trends, perspectives, and prospects. *Science* **2015**, *349*, 255–260. [CrossRef]
9. Kotsiantis, S.B.; Zaharakis, I.; Pintelas, P. Supervised Machine Learning: A Review of Classification Techniques. *Emerg. Artif. Intell. Appl. Comput. Eng.* **2007**, *160*, 2–24. [CrossRef]
10. Goodfellow, I.; Bengio, Y.; Courville, A.; RegGoodfellow, I.; Bengio, Y.; Courville, A. Regularization for Deep Learning. *Deep Learn.* **2016**, *1*, 216–261.
11. Bishop, C.M. Bishop-Pattern Recognition and Machine Learning-Springer 2006. *Antimicrob. Agents Chemother.* **2014**, *58*, 7250–7257.
12. Zhu, X. *Semi-Supervised Learning Literature Survey*; Technical Report 1530; Computer Sciences, University of Wisconsin-Madison: Madison, WI, USA, 2008; Available online: http://pages.cs.wisc.edu/~jerryzhu/pub/ssl_survey.pdf (accessed on 31 July 2025).
13. Chapelle, O.; Scholkopf, B.; Zien, A. Semi-Supervised Learning (Chapelle, O. et al., Eds.; 2006) [Book reviews]. *IEEE Trans. Neural Netw.* **2009**, *20*, 542. [CrossRef]
14. van Engelen, J.E.; Hoos, H.H. A survey on semi-supervised learning. *Mach. Learn.* **2020**, *109*, 373–440. [CrossRef]
15. Chen, Y.; Mancini, M.; Zhu, X.; Akata, Z. Semi-Supervised and Unsupervised Deep Visual Learning: A Survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **2024**, *46*, 1327–1347. [CrossRef]
16. Robert, C. Machine Learning, a Probabilistic Perspective. *Chance* **2014**, *27*, 62–63. [CrossRef]
17. Jain, A.K.; Murty, M.N.; Flynn, P.J. Data clustering: A review. *ACM Comput. Surv.* **1999**, *31*, 264–323. [CrossRef]
18. Jolliffe, I.T.; Cadima, J. Principal component analysis: A review and recent developments. *Philos. Trans. R. Soc. A Math. Phys. Eng. Sci.* **2016**, *374*, 20150202. [CrossRef]
19. Sutton, R.S.; Barto, A.G. Reinforcement Learning: An Introduction. *IEEE Trans. Neural Netw.* **2005**, *9*, 1054. [CrossRef]
20. Kaelbling, L.P.; Littman, M.L.; Moore, A.W. Reinforcement learning: A survey. *J. Artif. Intell. Res.* **1996**, *4*, 237–285. [CrossRef]
21. Szepesvári, C. *Algorithms for Reinforcement Learning*; Springer Nature: Berlin/Heidelberg, Germany, 2010. [CrossRef]

22. Bertsekas, D.P. Chapter 6 Approximate Dynamic Programming Approximate Dynamic Programming. In *Dynamic Programming and Optimal Control*, 3rd ed.; Athena Scientific: Belmont, MA, USA, 2010; Volume II.
23. Rasmussen, C.E. Gaussian Processes in machine learning. In *Advanced Lecture Notes in Computer Science*; Springer Nature: Berlin/Heidelberg, Germany, 2003; Volume 3176. [\[CrossRef\]](#)
24. Seeger, M. Gaussian processes for machine learning. *Int. J. Neural Syst.* **2004**, *14*, 69–106. [\[CrossRef\]](#)
25. Shahriari, B.; Swersky, K.; Wang, Z.; Adams, R.P.; De Freitas, N. Taking the human out of the loop: A review of Bayesian optimization. *Proc. IEEE* **2016**, *104*, 148–175. [\[CrossRef\]](#)
26. Williams, C.K.I. Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond. *J. Am. Stat. Assoc.* **2003**, *98*, 489. [\[CrossRef\]](#)
27. Cortes, C.; Vapnik, V. Support-Vector Networks. *Mach. Learn.* **1995**, *20*, 273–297. [\[CrossRef\]](#)
28. Freedman, D.A. *Statistical Models: Theory and Practice*; Cambridge University Press: Cambridge, UK, 2009. [\[CrossRef\]](#)
29. Scott, A.J.; Hosmer, D.W.; Lemeshow, S. Applied Logistic Regression. *Biometrics* **1991**, *47*, 1632–1633. [\[CrossRef\]](#)
30. Quinlan, J.R. Induction of Decision Trees. *Mach. Learn.* **1986**, *1*, 81–106. [\[CrossRef\]](#)
31. Breiman, L.; Friedman, J.H.; Olshen, R.A.; Stone, C.J. *Classification and Regression Trees*; Chapman and Hall: New York, NY, USA, 2017. [\[CrossRef\]](#)
32. Chen, T.; Guestrin, C. XGBoost: A scalable tree boosting system. In Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, New York, NY, USA, 13–18 August 2016. [\[CrossRef\]](#)
33. Powers, D.M.W. Evaluation: From precision, recall and f-measure to ROC, informedness, markedness & correlation. *arXiv* **2020**, arXiv:2010.16061.
34. Saito, T.; Rehmsmeier, M. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE* **2015**, *10*, e0118432. [\[CrossRef\]](#)
35. Chai, T.; Draxler, R.R. Root mean square error (RMSE) or mean absolute error (MAE)? -Arguments against avoiding RMSE in the literature. *Geosci. Model Dev.* **2014**, *7*, 1247–1250. [\[CrossRef\]](#)
36. Willmott, C.J.; Matsuura, K. Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance. *Clim. Res.* **2005**, *30*, 79–82. [\[CrossRef\]](#)
37. Hastie, T.; Tibshirani, R.; James, G.; Witten, D. *An Introduction to Statistical Learning*, 2nd ed.; Springer: New York, NY, USA, 2021; Volume 102.
38. Li, X.; Zhang, W.; Wang, F.; Sun, S.; Gu, C. Efficient parametric yield estimation of analog/mixed-signal circuits via Bayesian model fusion. In Proceedings of the IEEE/ACM International Conference on Computer-Aided Design, Digest of Technical Papers, ICCAD, San Jose, CA, USA, 4–9 November 2012. [\[CrossRef\]](#)
39. Wang, F.; Zaheer, M.; Li, X.; Plouchart, J.O.; Valdes-Garcia, A. Co-learning Bayesian model fusion: Efficient performance modeling of analog and mixed-signal circuits using side information. In Proceedings of the 2015 IEEE/ACM International Conference on Computer-Aided Design, ICCAD 2015, Austin, TX, USA, 2–6 November 2015. [\[CrossRef\]](#)
40. Wang, F.; Cachecho, P.; Zhang, W.; Sun, S.; Li, X.; Kanj, R.; Gu, C. Bayesian model fusion: Large-scale performance modeling of analog and mixed-signal circuits by reusing early-stage data. *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* **2016**, *35*, 1255–1268. [\[CrossRef\]](#)
41. Hasani, R.M.; Haerle, D.; Baumgartner, C.F.; Lomuscio, A.R.; Grosu, R. Compositional neural-network modeling of complex analog circuits. In Proceedings of the International Joint Conference on Neural Networks, Anchorage, AK, USA, 14–19 May 2017. [\[CrossRef\]](#)
42. Alawieh, M.; Wang, F.; Li, X. Efficient Hierarchical Performance Modeling for Integrated Circuits via Bayesian Co-Learning. In Proceedings of the 54th Annual Design Automation Conference, Austin, TX, USA, 18–22 June 2017. [\[CrossRef\]](#)
43. Yang, Y.; Zhu, H.; Bi, Z.; Yan, C.; Zhou, D.; Su, Y.; Zeng, X. Smart-MSP: A Self-Adaptive Multiple Starting Point Optimization Approach for Analog Circuit Synthesis. *IEEE Trans. Comput. Aided Des. Integr. Circuits Syst* **2018**, *37*, 531–544. [\[CrossRef\]](#)
44. Gao, Z.; Tao, J.; Yang, F.; Su, Y.; Zhou, D.; Zeng, X. Efficient performance trade-off modeling for analog circuit based on Bayesian neural network. In Proceedings of the IEEE/ACM International Conference on Computer-Aided Design, Digest of Technical Papers, ICCAD, Westminster, CO, USA, 4–7 November 2019. [\[CrossRef\]](#)
45. Islamoğlu, G.; Çakici, T.O.; Afacan, E.; Dundar, G. Artificial Neural Network Assisted Analog IC Sizing Tool. In Proceedings of the SMACD 2019—16th International Conference on Synthesis, Modeling, Analysis and Simulation Methods and Applications to Circuit Design, Lausanne, Switzerland, 15–18 July 2019. [\[CrossRef\]](#)
46. Hakhamaneshi, K.; Werblun, N.; Abbeel, P.; Stojanovic, V. BagNet: Berkeley analog generator with layout optimizer boosted with deep neural networks. In Proceedings of the IEEE/ACM International Conference on Computer-Aided Design, Digest of Technical Papers, ICCAD, Westminster, CO, USA, 4–7 November 2019. [\[CrossRef\]](#)
47. Zhang, S.; Lyu, W.; Yang, F.; Yan, C.; Zhou, D.; Zeng, X. Bayesian optimization approach for analog circuit synthesis using neural network. In Proceedings of the 2019 Design, Automation and Test in Europe Conference and Exhibition (DATE), Florence, Italy, 25–29 March 2019. [\[CrossRef\]](#)

48. Alawieh, M.B.; Williamson, S.A.; Pan, D.Z. Rethinking sparsity in performance modeling for analog and mixed circuits using spike and slab models. In Proceedings of the 56th Annual Design Automation Conference, Las Vegas, NV, USA, 2–6 June 2019. [\[CrossRef\]](#)
49. Alawieh, M.B.; Tang, X.; Pan, D.Z. S²-PM: Semi-supervised learning for efficient performance modeling of analog and mixed signal circuits. In Proceedings of the 24th Asia and South Pacific Design Automation Conference, ASP-DAC, Tokyo, Japan, 21–24 January 2019. [\[CrossRef\]](#)
50. Gabourie, A.; McClellan, C.; Suryavanshi, S. Analog Circuit Design Enhanced with Artificial Intelligence. 2022. Available online: <https://web.stanford.edu/class/archive/cs/cs221/cs221.1192/2018/restricted/posters/cjmcc/poster.pdf> (accessed on 31 July 2025).
51. Wang, M.; Lv, W.; Yang, F.; Yan, C.; Cai, W.; Zhou, D.; Zeng, X. Efficient Yield Optimization for Analog and SRAM Circuits via Gaussian Process Regression and Adaptive Yield Estimation. *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* **2018**, *37*, 1929–1942. [\[CrossRef\]](#)
52. Lyu, W.; Xue, P.; Yang, F.; Yan, C.; Hong, Z.; Zeng, X.; Zhou, D. An efficient Bayesian optimization approach for automated optimization of analog circuits. *IEEE Trans. Circuits Syst. I Regul. Pap.* **2018**, *65*, 1954–1967. [\[CrossRef\]](#)
53. Lyu, W.; Yang, F.; Yan, C.; Zhou, D.; Zeng, X. Batch Bayesian optimization via multi-objective acquisition ensemble for automated analog circuit design. In Proceedings of the 35th International Conference on Machine Learning, ICML 2018, Stockholm, Sweden, 10–15 July 2018.
54. Pan, P.C.; Huang, C.C.; Chen, H.M. Late breaking results: An efficient learning-based approach for performance exploration on analog and RF circuit synthesis. In Proceedings of the 56th Annual Design Automation Conference, Las Vegas, NV, USA, 2–6 June 2019. [\[CrossRef\]](#)
55. Li, Y.; Wang, Y.; Li, Y.; Zhou, R.; Lin, Z. An Artificial Neural Network Assisted Optimization System for Analog Design Space Exploration. *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* **2020**, *39*, 2640–2653. [\[CrossRef\]](#)
56. Zhao, Z.; Zhang, L. Deep reinforcement learning for analog circuit sizing. In Proceedings of the IEEE International Symposium on Circuits and Systems, Seville, Spain, 12–14 October 2020. [\[CrossRef\]](#)
57. Wang, H.; Yang, J.; Lee, H.-S.; Han, S. Learning to Design Circuits. *arXiv* **2020**, arXiv:1812.02734. [\[CrossRef\]](#)
58. Devi, S.; Tilwankar, G.; Zele, R. Automated Design of Analog Circuits using Machine Learning Techniques. In Proceedings of the 2021 25th International Symposium on VLSI Design and Test, (VDAT), Surat, India, 16–18 September 2021. [\[CrossRef\]](#)
59. Yin, Y.; Wang, Y.; Xu, B.; Li, P. ADO-LLM: Analog design bayesian optimization with in-context learning of large language models. In Proceedings of the IEEE/ACM International Conference on Computer-Aided Design, Digest of Technical Papers, ICCAD, New York, NY, USA, 27–31 October 2024. [\[CrossRef\]](#)
60. Dumesnil, E.; Nabki, F.; Boukadoum, M. RF-LNA circuit synthesis by genetic algorithm-specified artificial neural network. In Proceedings of the 2014 21st IEEE International Conference on Electronics, Circuits and Systems, (ICECS), Marseille, France, 7–10 December 2014. [\[CrossRef\]](#)
61. Fukuda, M.; Ishii, T.; Takai, N. OP-AMP sizing by inference of element values using machine learning. In Proceedings of the 2017 International Symposium on Intelligent Signal Processing and Communication Systems, (ISPACS), Xiamen, China, 6–9 November 2017. [\[CrossRef\]](#)
62. Lourenço, N.; Rosa, J.; Martins, R.; Aidos, H.; Canelas, A.; Póvoa, R.; Horta, N. On the exploration of promising analog ic designs via artificial neural networks. In Proceedings of the SMACD 2018—15th International Conference on Synthesis, Modeling, Analysis and Simulation Methods and Applications to Circuit Design, Prague, Czech Republic, 2–5 July 2018. [\[CrossRef\]](#)
63. Wang, Z.; Luo, X.; Gong, Z. Application of deep learning in analog circuit sizing. In Proceedings of the ACM International Conference Proceeding Series, Shenzhen, China, 8–10 December 2018. [\[CrossRef\]](#)
64. Lourenço, N.; Afacan, E.; Martins, R.; Passos, F.; Canelas, A.; Póvoa, R.; Horta, N.; Dundar, G. Using polynomial regression and artificial neural networks for reusable analog IC sizing. In Proceedings of the SMACD 2019—16th International Conference on Synthesis, Modeling, Analysis and Simulation Methods and Applications to Circuit Design, Lausanne, Switzerland, 15–18 July 2019. [\[CrossRef\]](#)
65. Harsha, V.M.; Harish, B.P. Artificial neural network model for design optimization of 2-stage Op-amp. In Proceedings of the 2020 24th International Symposium on VLSI Design and Test, (VDAT), Bhubaneswar, India, 23–25 July 2020. [\[CrossRef\]](#)
66. Murphy, S.D.; McCarthy, K.G. Automated design of CMOS operational amplifier using a neural network. In Proceedings of the 2021 32nd Irish Signals and Systems Conference, (ISSC), Athlone, Ireland, 10–11 June 2021. [\[CrossRef\]](#)
67. Zhang, J.; Huang, L.; Wang, Z.; Verma, N. A seizure-detection IC employing machine learning to overcome data-conversion and analog-processing non-idealities. In Proceedings of the 2015 IEEE Custom Integrated Circuits Conference (CICC), San Jose, CA, USA, 28–30 September 2015. [\[CrossRef\]](#)
68. Hakhamaneshi, K.; Werblun, N.; Abbeel, P.; Stojanović, V. Late breaking results: Analog circuit generator based on deep neural network enhanced combinatorial optimization. In Proceedings of the 56th Design Automation Conference, Las Vegas, NV, USA, 2–6 June 2019. [\[CrossRef\]](#)

69. Basso, D.; Bortolussi, L.; Videnovic-Misic, M.; Habal, H. Fast ML-driven Analog circuit layout using reinforcement learning and steiner trees. In Proceedings of the 2024 20th International Conference on Synthesis, Modeling, Analysis and Simulation Methods and Applications to Circuit Design (SMACD), Volos, Greece, 2–5 July 2024. [\[CrossRef\]](#)
70. Mehradfar, A.; Zhao, X.; Niu, Y.; Babakniya, S.; Alesheikh, M.; Aghasi, H.; Avestimehr, S. AICircuit: A multi-level dataset and benchmark for ai-driven analog integrated circuit design. *arXiv* **2024**, arXiv:2407.18272. [\[CrossRef\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.